

Review article

What modulates the Mirror Neuron System during action observation? Multiple factors involving the action, the actor, the observer, the relationship between actor and observer, and the context

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ABSTRACT

Seeing an agent perform an action typically triggers a motor simulation of that action in the observer's Mirror Neuron System (MNS). Over the past few years, it has become increasingly clear that during action observation the patterns and strengths of responses in the MNS are modulated by multiple factors. The first aim of this paper is therefore to provide the most comprehensive survey to date of these factors. To that end, 22 distinct factors are described, broken down into the following sets: six involving the action; two involving the actor; nine involving the observer; four involving the relationship between actor and observer; and one involving the context. The second aim is to consider the implications of these findings for four prominent theoretical models of the MNS: the Direct Matching Model; the Predictive Coding Model; the Value-Driven Model; and the Associative Model. These assessments suggest that although each model is supported by a wide range of findings, each one is also challenged by other findings and relatively unaffected by still others. Hence, there is now a pressing need for a richer, more inclusive model that is better able to account for all of the modulatory factors that have been identified so far.

1. Introduction

In our daily lives, when we see other people perform ordinary actions like grasping and manipulating objects, walking and running, or eating and drinking, our brains usually simulate those actions in a covert, unconscious manner. This phenomenon, which is sometimes called motor resonance, first came to light with the discovery of mirror neurons in macaque monkeys nearly 30 years ago (di Pellegrino et al., 1992; Gallese et al., 1996; Rizzolatti et al., 1996), and ever since then it has been the focus of intense empirical and theoretical research. Mirror activity is widespread in the human brain (Gazzola and Keysers, 2009), but the core of the human Mirror Neuron System (MNS) is generally thought to consist of cortical areas located bilaterally in the inferior parietal lobule (especially the supramarginal gyrus and adjacent intraparietal sulcus), the ventral premotor cortex, and the posterior inferior frontal gyrus; in addition, the most prominent components of the extended MNS are the primary motor cortex and supplementary motor area (Fig. 1; Van Overwalle and Baetens, 2009; Caspers et al., 2010; Fogassi and Ferrari, 2011; Molenberghs et al., 2012; Avenanti et al., 2013; Urgesi et al., 2014; Rizzolatti et al., 2014; Bonini, 2017; Hardwick et al., 2018).

During the past few years, accumulating data from diverse brain

mapping methods—fMRI, positron emission tomography (PET), transcranial magnetic stimulation (TMS), magnetoencephalography (MEG), intracranial and extracranial electrophysiology, and lesions—has made it increasingly clear that during action observation the patterns and strengths of responses in the MNS are modulated by multiple factors. Several recent papers have begun to take stock of these discoveries and evaluate their repercussions (Naish et al., 2014; Campbell and Cunningham, 2017; Aziz-Zadeh et al., 2018; Amoroso and Finisguerra, 2019; Finisguerra et al., 2020). However, none of these papers has addressed the full range of factors that have been found to differentially affect the MNS. The first and primary aim of this paper is therefore to provide a much broader review of this topic.

A useful way to organize and compare the relevant factors is to classify them according to whether they predominantly involve the action itself, the actor, the observer, the relationship between actor and observer, or the context. This approach is illustrated in Fig. 2, and several noteworthy aspects of the proposed taxonomy are as follows. First, although this inventory of factors is not comprehensive, it does include the vast majority of factors that have been found to differentially affect the MNS. Second, a few factors could, in principle, be broken down further, but the arrangement adopted here is nonetheless well-

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justified. For example, the topics lumped together in the observer-related factor called "Motivation, emotion, personality, and moral judgment" could be treated separately. However, as discussed in greater detail below, all of these topics are closely related insofar as they involve subjective value-based influences on the MNS. Conversely, a few factors could, in principle, be integrated into a single, more general one, but again the current approach is defensible. For example, it will become clear that the factors called "Perspective," "Handedness," and "Postural congruence" have much in common. However, they also have nontrivial differences, and furthermore they have, for the most part, been studied independently in the literature. Another important point is that even though the first aim of this paper is to bring together the lion's share of factors that modulate the MNS, the available literature is simply far too large to review in detail here. Hence, for those factors that have been extensively studied, only the most relevant results are highlighted, thereby preserving enough space to cover less familiar factors too. Finally, due to the large number of studies that are summarized, emphasis is placed on concise description rather than critical commentary.¹

The fact that motor resonance during action observation is not an all-or-nothing affair, but is instead modulated by myriad factors, has significant theoretical implications. Another aim of this paper is therefore to explore how well four prominent models of the MNS can account for the various findings. These frameworks are as follows: the Direct Matching Model (Rizzolatti et al., 2001, 2014; Rizzolatti and Sinigaglia, 2010); the Predictive Coding Model (Kilner et al., 2007a, 2007b); the Value-Driven Model (Aziz-Zadeh et al., 2018); and the Associative Model (Heyes, 2010; Cook et al., 2014; Catmur et al., 2016). (For other recent approaches, see the Discussion Forum called "The meaning of mirror neurons" in *Cortex*, volume 49, 2013. See also D'Ausilio et al. (2015); Amoroso and Finisguerra (2019); Finisguerra et al. (2020)). It will be argued that each model is supported by some findings, challenged by others, and largely unaffected by still others.

The brain regions comprising the MNS are, by definition, typically engaged during both the execution and the observation of the same kinds of actions. However, while many studies have confirmed that the regions shown in Fig. 1 satisfy both criteria, in some cases researchers have inferred MNS recruitment even though they only examined action observation. Due to the large volume of studies that are considered in this paper, no attempt was made to indicate for each one whether it

investigated both action execution and action observation or only the latter, since doing so would be stylistically tedious. Nonetheless, there are numerous instances in which it is clear that both execution and observation were examined.

The paper is organized such that the review of the multifarious empirical findings precedes the discussion of their theoretical implications. More specifically, the first five sections focus on the five categories of modulatory factors listed in Fig. 2: factors involving the action; factors involving the actor; factors involving the observer; factors involving the relationship between actor and observer; and the factor of context. The next section then shifts to the four models mentioned above. Each model is addressed in turn, with an initial description of its central tenets, followed by an assessment of its ability to account for the various factors. (An alternative approach would be to describe the models before discussing the various factors, and then track their relative strengths and weaknesses in parallel with the review of empirical findings. But the organizational strategy adopted here is arguably more economical, since it affords more holistic evaluations and comparisons of the models. Readers who would nonetheless like to see these treatments in advance should consult Table 1 and the associated text in the section called "Theoretical implications"). Finally, the concluding section summarizes the key points of the paper and underscores what may be its most significant message—namely, that theoretical work on the MNS has been lagging behind empirical work, and there is now a pressing need for a richer, more inclusive model that is better able to account for the full range of modulatory factors that have been identified so far.

2. Factors involving the action

2.1. Factor 1: transitivity (i.e., +/- object-directed)

One of the first factors that was found to modulate the MNS is transitivity. To clarify the key distinction, transitive actions are directed toward objects in peripersonal space (e.g., grasping a mug), whereas intransitive actions are not (e.g., moving an index finger up and down). According to several influential review articles, mirror neurons in monkeys respond to transitive but not intransitive hand actions (Rizzolatti et al., 2001, 2014; Rizzolatti and Craighero, 2004; Rizzolatti and Sinigaglia, 2010). One recent study, however, did obtain compelling evidence that some mirror neurons in monkeys are sensitive to intran-



Fig. 1. Parietal and frontal components of the human Mirror Neuron System (MNS). The core MNS is generally thought to include the following cortical regions: the inferior parietal lobule, especially the supramarginal gyrus and adjacent intraparietal sulcus (yellow); the ventral premotor cortex (green); and the posterior inferior frontal gyrus (red). However, the extended MNS contains additional regions, including the primary motor cortex (blue) and supplementary motor area (magenta). Although these structures are only shown here in the left hemisphere (lateral view on the left, medial view on the right), the homologous structures in the right hemisphere are also implicated.

¹ This paper only covers the normal operation of the MNS; it does not address the effects corporeal, neurological, and psychiatric disorders. For research on the status of the MNS in people born without arms or hands, see the following fMRI studies: Gazzola et al. (2007b); Aziz-Zadeh et al. (2012); and Vannucorps et al. (2019). For research on the status of the MNS in people with neurodegenerative disorders, see the following review paper: Farina et al. (2019). And for research on the status of the MNS in people with autism, see the following review papers: Hamilton (2013); and Kilroy et al. (2019).

sitive hand actions (Papadourakis and Raos, 2017). Turning to humans, copious studies have used PET, fMRI, TMS, and electroencephalography (EEG) to examine the impact of transitivity on action observation, and there is now a voluminous amount of data on this topic. Due to space limitations, however, the following summary is restricted to the most relevant findings.

Meta-analyses of PET and fMRI studies have shown that people typically display greater MNS responses to transitive than intransitive hand actions, especially in frontal regions, but also in parietal ones

(Caspers et al., 2010). In addition, at least one TMS study found that the primary motor cortex is sensitive to the sight of transitive but not intransitive hand actions (Enticott et al., 2010). These results suggest that the MNS is significantly influenced by transitivity, with stronger and more widely distributed responses to object-directed than non-object-directed behaviors. Notably, however, all of the findings just mentioned are restricted to hand actions, and the meta-analytic PET/fMRI findings only reflect probabilistic trends across disparate investigations.

In fact, many other studies have shown that the observation of intransitive actions does evoke MNS responses in parietal and frontal regions that overlap partly with, but are also partly segregated from, those that respond to transitive actions. In the fMRI literature, for instance, such results have been reported for intransitive actions that involve simple finger movements (e.g., Iacoboni et al., 1999), complex finger movements (e.g., Plata Bello et al., 2013), simple movements of the mouth, knee, ankle, elbow, wrist, trunk, and shoulder (e.g., Sakreida et al., 2005), and complex leg movements such as walking (e.g., Dalla Volta et al., 2015) and dancing (e.g., Calvo-Merino et al., 2005). Moreover, starting with the seminal TMS study by Fadiga et al. (1995), numerous TMS investigations have shown that the primary motor cortex mirrors intransitive actions in a fine-grained manner (for a review see Naish et al., 2014). And EEG studies have provided further evidence that several components of the MNS are engaged by the sight of both transitive and intransitive hand actions, with somewhat different responses to these two types of stimuli (e.g., Arnstein et al., 2011).

intransitive action. Picking up on the latter usage, one potentially important consideration is that perceived actions can be coded in the onlooker's MNS in a flexible fashion that emphasizes any of several different levels of representation, from low-level muscle commands up to high-level intentions (for a review see Amoruso and Finisguerra, 2019). Although intransitive actions are, by definition, not directed toward objects, they are usually performed on purpose; hence, they may sometimes evoke MNS responses not only to their purely physical form, but also to their underlying aim. This is especially likely when that aim can be inferred, as with intransitive actions that have communicative functions, such as symbolic gestures (e.g., "thumbs-up") and spoken words. Indeed, the sight of such stimuli does engage many of the same parietal and frontal regions that respond to the sight of transitive grasping actions (Andric et al., 2013; Maffei et al., 2020). Furthermore, in a particularly interesting fMRI study, Möttönen et al. (2016) showed participants unfamiliar symbolic gestures and found that when the participants assumed the gestures were just meaningless intransitive actions, the strongest MNS response was in the premotor cortex, presumably reflecting physical features of the movements; however, when the participants knew that the gestures were communicative, but didn't yet know their actual meanings, the left posterior inferior frontal gyrus (pIFG) was recruited as well, supporting the view that this region plays a special role in the attribution of intention² (see also Levänen et al., 2011; MacSweeney et al., 2004, and the discussions of action purposes in the subsections on *Factors 2* and *15*). More generally, it seems that the intentionality of intransitive actions is one property that influences the

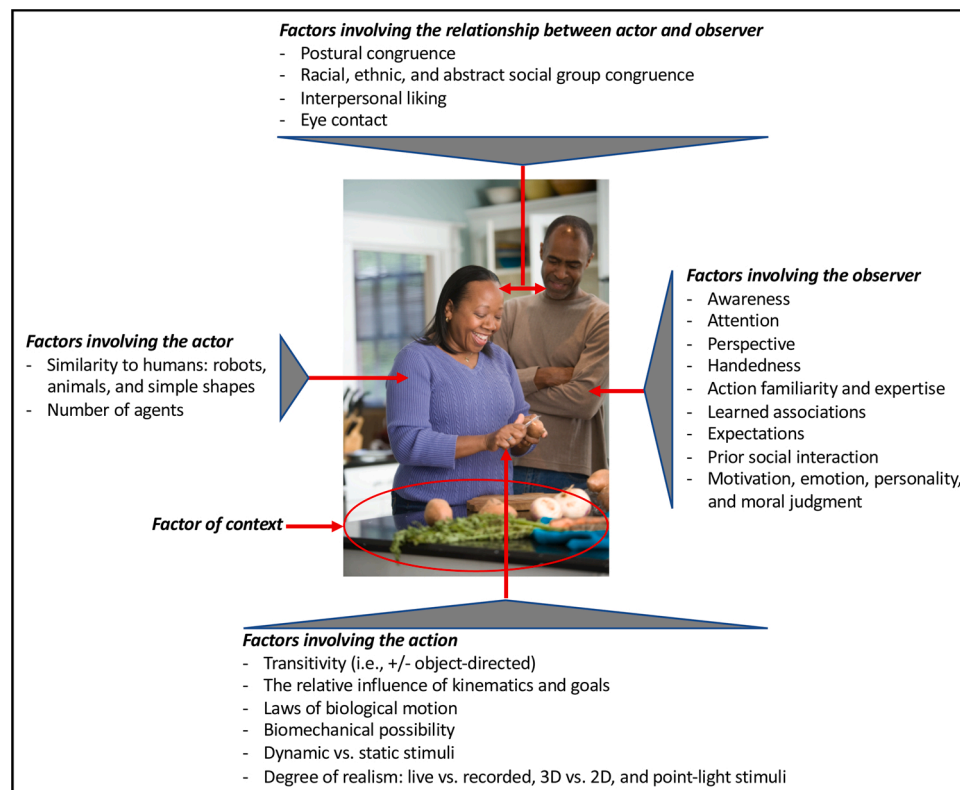


Fig. 2. Inventory of factors that modulate the MNS during action observation, grouped into five categories: factors involving the action itself; factors involving the actor; factors involving the observer; factors involving the relationship between actor and observer; and the factor of context. Image source: <https://unsplash.com/s/photos/cooking>.

Thus, it is apparent that transitivity does modulate the MNS, but it is not yet clear exactly how. This is partly because the term "goal" has been used inconsistently in much of the literature, sometimes referring to the object that is the spatial endpoint of a transitive action, and sometimes referring to an agent's reason for executing either a transitive or an

² An anonymous reviewer pointed out, however, that the pIFG engagement could reflect the processing of communicatively relevant manual movements rather than the more abstract intention to communicate.

Table 1

Assessment of how the factors that modulate the MNS have implications for four models of the MNS. The following codes are used: "+" indicates that the evidence regarding the given factor supports the given model; "-" indicates that the evidence challenges the model; "o" indicates that the evidence does not bear strongly on the model; and "x" indicates that the evidence is insufficient to evaluate the model. As described in the main text, some of these judgments are only tentative because the precise predictions of the models are not always clear. Abbreviations: DMM, Direct Matching Model; PCM, Predictive Coding Model; VDM, Value-Driven Model; AM, Associative Model.

Factors	DMM	PCM	VDM	AM
<i>Factors involving the action</i>				
1. Transitivity (i.e., +/- object-directed)	+	+	+	+
2. The relative influence of kinematics and goals	+	+	+	+
3. Laws of biological motion	+	+	+	+
4. Biomechanical possibility	x	x	x	x
5. Dynamic vs. static stimuli	-	+	+	+
6. Degree of realism: live vs. recorded, 3D vs. 2D, & point-light stimuli	o	o	+	o
<i>Factors involving the actor</i>				
7a. Similarity to humans: robots	x	x	x	x
7b. Similarity to humans: animals	+	+	+	+
7c. Similarity to humans: simple shapes	-	+	+	+
8. Number of agents	+	o	o	+
<i>Factors involving the observer</i>				
9. Awareness	o	o	o	o
10. Attention	o	o	o	o
11. Perspective	-	o	o	o
12. Handedness	-	o	o	+
13a. Action familiarity/expertise: positive relation to mirror responses	+	+	+	+
13b. Action familiarity/expertise: negative relation to mirror responses	-	+	+	-
14a. Learned associations: development of counter-mirror responses	-	o	o	+
14b. Learned associations: preservation of normal mirror responses	+	o	o	-
15a. Expectations: sometimes do influence mirror responses	-	+	+	+
15b. Expectations: sometimes do not influence mirror responses	+	-	-	-
16. Prior social interaction	o	o	o	o
17. Motivation, emotion, personality, and moral judgment	-	o	+	o
<i>Factors involving the relationship between actor and observer</i>				
18. Postural congruence	+	o	o	o
19. Racial, ethnic, and abstract social group congruence	-	o	+	o
20. Interpersonal liking	-	o	+	o
21. Eye contact	-	o	+	o
22. The factor of context	-	+	+	+

degree to which they trigger MNS responses that are similar to those triggered by transitive actions. Further research is needed, though, to understand this and other ways in which transitivity modulates the MNS.

2.2. Factor 2: the relative influence of kinematics and goals

There is compelling evidence that the MNS is sensitive by both the kinematics (i.e., the changing postures and trajectories of body parts) and the goals (i.e., the target objects and overarching purposes) of observed actions. However, there has also been some controversy regarding the relative influence of these properties, and recent research suggests that they have different effects depending on the experimental design.

Evidence that the primary motor cortex represents the fine-grained kinematics of observed actions comes from many sources, but especially from Naish et al. (2014), who reviewed many TMS studies on this topic and drew two conclusions: first, action observation induces an early facilitation of corticospinal excitability that begins about 90 ms from action onset but is not yet muscle-specific; and second, this is

followed by a later phase of facilitation that starts around 200 ms and is muscle-specific. Importantly, the later phase often involves tightly coupled observation-execution matching, such that the temporally unfolding pattern of corticospinal excitability in the onlooker replicates the progressive changes of muscle activity in the agent. The primary motor cortex even mimics extremely subtle alterations of observed grasping movements (Senot et al., 2011; Sartori et al., 2012), and reducing its excitability, specifically by means of continuous theta burst stimulation, interferes with subjects' discrimination of such movements (Palmer et al., 2016).

During action observation, the muscle-specific response patterns in the primary motor cortex appear to be orchestrated by the pIFG (especially Brodmann area [BA] 44), which is also sensitive to perceived kinematics and widely regarded as a major component of the MNS (Nishitani and Hari, 2000). For example, an fMRI study by Kilner et al. (2009) suggests that this region houses neural populations that represent particular types of hand actions, such as a precision grip or an index finger pull, regardless of whether they are executed or observed, much like the strictly congruent mirror neurons that were discovered in the macaque monkey's area F5, which may be the homolog of BA44 in the human pIFG (see also Hamilton and Grafton, 2008b; Hesse et al., 2008). In addition, TMS studies have shown that temporarily disrupting the pIFG impairs subjects' ability to not only judge the weight of a box lifted by an agent (Pobric and Hamilton, 2006) but also distinguish between two images of slightly different body postures (Urgesi et al., 2007)—tasks that arguably require the processing of kinematics.

Mirror responses in parietal and frontal regions are also clearly influenced by the goals of observed actions. When these goals are defined as just the target objects of reaching and grasping actions, they are represented, independent of kinematics, in the anterior intraparietal sulcus (Shmuelof and Zohary, 2006; Hamilton and Grafton, 2006, 2008b; Tunik et al., 2007). However, when the goals of observed actions are defined not in terms of target objects but rather in terms of the agent's more abstract purposes (e.g., to open or close a container), both parietal and frontal components of the MNS are recruited. Although some fMRI studies have only implicated parietal areas in this relatively high level of action representation (Jastorff et al., 2010; Wurm and Lingnau, 2015; Wurm et al., 2016; Orban et al., 2019), others have also implicated several frontal areas, most reliably the pIFG (Gazzola et al., 2007a; de Lange et al., 2008; Hamilton and Grafton, 2008a; Andric et al., 2013; Möttönen et al., 2016; Koul et al., 2018). Further support for a critical role of the left pIFG comes from a neuropsychological study which found that in a group of stroke patients with limb apraxia, this region was the site of greatest lesion overlap for those patients who were impaired on a gesture recognition task that "assessed the ability to judge whether the ultimate goals of transitive gestures are attained or whether the symbolic meaning of intransitive gestures is maintained" (Pazzaglia et al., 2008, p. 3039; see also the repetitive TMS experiment by Tidoni et al., 2013, which is discussed in the subsection on Factor 15).

As mentioned above, there has also been some debate over the relative influence of kinematics and goals on MNS responses. Much of this controversy has focused on the engagement of the primary motor cortex during the observation of actions involving tool use. In a TMS study Cattaneo et al. (2009) measured the corticospinal excitability of the opponens pollicis (OP) muscle, which controls thumb movement, while subjects watched an agent grasp objects with either classic pliers, which are closed with a closing hand movement, or reverse pliers, which are closed with an opening hand movement. They found that the pattern of OP excitability corresponded best to the closing movements of the distal parts of both kinds of pliers, which they interpreted as evidence that motor resonance was driven by the common goals of the actions—namely, object grasping. Subsequently, however, Cavallo et al. (2012) pointed out that this interpretation conflicts with a large literature showing that the primary motor cortex is typically not sensitive to the ultimate effects of observed actions, but instead tends to track the kinematics of such actions in a time-locked, muscle-specific manner. To

fully dissociate hand movements, tool movements, and action outcomes, these researchers conducted another TMS study in which they measured the corticospinal excitability of both the OP and the first dorsal interosseous (FDI), which controls index finger movement, while subjects watched grasping actions performed not only with classic pliers and reverse pliers, but also with magnetic pliers, which are opened with an opening hand movement and then maintained in that state. As in [Cattaneo et al.'s \(2009\)](#) study, the pattern of OP excitability was similar for the actions involving both classic and reverse pliers; however, the pattern of FDI excitability was tightly coupled with the agent's use of that particular muscle, mirroring the dynamics of index finger movements across all three conditions. Cavallo et al. (2012, p. 714) therefore drew the following conclusion: "In contrast to the hypothesis that [motor] facilitation reflects the action goal regardless of the movements necessary to achieve it, this finding suggests that [motor] facilitation reflects the observed hand movements".

Further complications, however, come from TMS studies by [Senna et al. \(2014\)](#) and [Finisguerra et al. \(2015\)](#). In [Senna et al.'s \(2014\)](#) study, subjects observed grasping actions performed with either a typical body part—namely, a hand—or an atypical body part—namely, a foot. Interestingly, the foot condition enhanced corticospinal excitability not only for a relevant foot muscle but also for a relevant hand muscle, reflecting covert motor simulation of both the peculiar action that was actually seen and the more customary action that was not seen. These results suggest that the primary motor cortex may implement what [Senna et al. \(2014, p. 1750\)](#) call "a double coding of observed actions: a strict somatotopic coding and an action goal coding based on the observer's motor expertise". Relatedly, [Finisguerra et al. \(2015\)](#) found that when subjects observed intransitive closing hand movements, corticospinal excitability was facilitated not only for a forearm flexor muscle that supports the execution of such movements, but also for eye and mouth muscles that can perform similar movements at the goal level of "closing". (See also [Amoruso & Urgesi's \(2016\)](#) finding that, as described in connection with *Factor 7*, the sight of a dog grasping an object with its mouth increased the corticospinal excitability of an arm/wrist muscle in observers who had substantial familiarity with dogs).

2.3. Factor 3: laws of biological motion

Human actions exhibit certain kinematic regularities that reflect natural laws of biological motion, and a few fMRI studies suggest that these laws also modulate the MNS during action observation.

One invariant is that when people perform actions with their arms, hands, legs, feet, head, etc., there is a close relationship between velocity and curvature, such that movements slow down during more curved segments and speed up during straighter ones, thereby maximizing smoothness and precision. This relationship can be expressed in a mathematically rigorous way and is known as the two-thirds power law. Neurophysiological studies with monkeys have found the imprint of this law in the activation dynamics of cells in the premotor and primary motor cortices (Schwartz and Moran, 2000). Moreover, in an fMRI study [Casile et al. \(2010\)](#) presented people with computer-generated videos of intransitive rotational arm movements and found that, relative to movements that violated the two-thirds power law, those that complied with it induced greater activation in left premotor and dorsofrontal regions (see also [Dayan et al., 2007](#)). This discovery suggests that kinematically natural actions trigger stronger MNS responses than kinematically unnatural ones (for further support see [Shimada, 2010](#), and [Press et al., 2011](#); but note that contrary findings were reported by [Cross et al., 2012](#), and [Craighero et al., 2016](#)).

More recently, in another fMRI study [Cignetti et al. \(2017\)](#) showed subjects a variety of everyday actions, like crouching, tilting forward or backward, and standing up from or sitting down onto a chair. Crucially, however, some of the actions were executed in an environment with normal gravity, whereas others were executed in an environment with almost no gravity. The task was to distinguish between these two

conditions, and subjects did so successfully. Even more interesting, though, is that the observation of normogravity actions engaged the right primary motor cortex significantly more than the observation of microgravity actions. This result provides additional evidence that actions with ordinary movement patterns are more likely to be mapped onto the viewer's own motor system than actions with strange ones.

2.4. Factor 4: biomechanical possibility

This factor is closely related to the previous one. The key question is how the MNS reacts to the sight of biomechanically impossible actions that, by definition, the observer could not perform and hence could not simulate precisely either. Although several studies have addressed this issue, the results are mixed.

In the first investigation, [Stevens et al. \(2000\)](#) used PET to compare the activation patterns engendered by different kinds of illusory human motion patterns. They found that although parietal and frontal components of the MNS were engaged when subjects saw what appeared to be a man moving his arm around his leg (a possible action), the same regions were not engaged when subjects saw what appeared to be a man moving his arm directly through his leg (an impossible action).

Different results, however, were obtained by subsequent studies that used possible and impossible finger movements as stimuli. For one thing, an fMRI study by [Constantini et al. \(2005\)](#) revealed that impossible finger movements activated somatosensory regions more than possible ones—a discovery that matched the subjects' reports that the former stimuli evoked feelings of discomfort (see also [Avenanti et al., 2007](#)). More interestingly, two TMS studies focused on the primary motor cortex and found that, remarkably enough, the observation of both possible and impossible finger movements facilitated the corticospinal excitability of pertinent muscles ([Romani et al., 2005](#); [Borroni et al., 2011](#)). As [Romani et al. \(2005\)](#) point out, the primary motor cortex may have represented the impossible movements as exaggerations of the possible ones, or as unusual actions that could potentially be learned. In any event, the results suggest that "excitation of the human corticospinal system contingent upon action observation is related more to coding of the specificity of moving body parts than to a detailed reading of the kinematics of the observed movement in terms of its biological plausibility" ([Romani et al., 2005, p. 761](#)).

The ventral premotor cortex is also relevant here, but the data are rather puzzling. The fMRI study by [Constantini et al. \(2005\)](#) found that this region was equally engaged by possible and impossible finger movements; however, repetitive TMS studies by [Avenanti et al. \(2007\)](#) and [Candidi et al. \(2008\)](#) found that "virtual lesions" of the ventral premotor cortex only disrupted the ability to discriminate between possible finger movements, leaving intact the ability to discriminate not only between impossible finger movements, but also between possible and impossible ones. According to [Candidi et al. \(2008, p. 398\)](#), one way to account for these results would be to assume that the ventral premotor cortex codes "a combination of goals and kinematics that allow the representation of actions as goal-directed movements ([Rizzolatti and Craighero, 2004](#))". Further research is needed, however, to better understand not only how biomechanical possibility modulates the MNS, but also how this factor interacts with other closely related ones (see especially *Factor 3* and the discussion of rigid robotic movements in the subsection on *Factor 7*).

2.5. Factor 5: dynamic vs. static stimuli

An interesting question that has drawn some attention is whether the MNS is affected by whether actions are presented as genuine dynamic events or just static snapshots. It's worth beginning with the notion of "representational momentum" ([Kelly and Freyd, 1987](#)). This is a well-established phenomenon whereby observers of static action stimuli impulsively extrapolate motion forward in time to the next phase—e.g., anticipating the next limb movements of a runner when shown a photo

of her in mid-stride. The robustness of representational momentum suggests that perceiving and understanding static depictions of familiar actions depends on predictive processing derived from previous experience and conceptual knowledge. The question here is whether the MNS contributes to such processing, and if so, whether it is engaged by static action stimuli as much as it is by dynamic action stimuli.

Evidence that static action stimuli do engage the MNS comes from several sources (for connections with art appreciation see [Freedberg and Gallese, 2007](#)). For example, fMRI studies have shown that both parietal and frontal components of the MNS are enlisted when subjects observe not only stationary images of object-directed hand movements ([Hamzei et al., 2003](#); [Chong et al., 2008](#)), but also action pictograms, which are schematic stick-figure-like portrayals of complex human movements like golfing, skating, and diving ([Assmus et al., 2007](#); [Quandt et al., 2017](#)). Another fMRI study found that when subjects judged sequences of three static body postures, responses in primary and supplementary motor areas were greater when those sequences were natural and fluent (ABC) than when they were unnatural and nonfluent (ACB) ([Orgs et al., 2016](#)). In addition, a number of neuropsychological and TMS studies have shown that real or virtual lesions to left inferior frontal, premotor, primary motor, and/or inferior parietal regions impair performance on nonlinguistic tasks that require judgments about static action stimuli ([Tranel et al., 2003](#); [Bak and Hodges, 2004](#); [Saygin et al., 2004](#); [Hillis et al., 2006](#); [Urgesi et al., 2007](#); [Moro et al., 2008](#); [Fazio et al., 2009](#); [Kemmerer et al., 2012](#)). Furthermore, a TMS study showed that, compared with static images of either a relaxed hand or a hand at the endpoint of an action, static images of a hand executing a thumb-forefinger pincer grip increased the excitability of the muscles that would perform that action; those muscles were not affected, however, by static images of nonbiological entities with (e.g., waterfalls) or without (e.g., icefalls) implied motion ([Urgesi et al., 2006](#); for follow-up studies see [Urgesi et al., 2010](#), and [Mattiassi et al., 2014](#); and for contrary data see [Bunday et al., 2016](#)). Finally, an event-related potential (ERP) study showed that, relative to static photos of less dynamic actions, static photos of highly dynamic ones triggered a positivity 350–600 ms post-stimulus, traceable to a set of movement-related brain regions that included premotor and primary motor cortices ([Proverbio et al., 2009](#)).

Although the results reviewed above suggest that static action stimuli do trigger predictive motor simulations in the observer's brain, they don't indicate whether such MNS responses are as strong as those triggered by dynamic action stimuli. This issue has not yet been carefully investigated, but a recent fMRI study found that the MNS responses evoked by static action stimuli were weaker than those evoked by dynamic action stimuli ([Ferri et al., 2015](#)).

2.6. Factor 6: degree of realism: live vs. recorded, 3D vs. 2D, and point-light stimuli

Another factor that modulates the MNS is the degree to which the action can be perceived as a real physical occurrence. It's worth beginning with a study involving macaque monkeys which found that of 123 mirror neurons in area F5, 53 (43 %) fired more frequently in response to naturalistic than filmed actions ([Caggiano et al., 2011](#)). Turning to humans, a MEG study found that the primary motor cortex was engaged significantly more during the observation of live than videotaped hand movements ([Järveläinen et al., 2001](#); see also [Ruyschaert et al., 2013](#)). Similarly, a TMS study found that the corticospinal excitability of an arm muscle was significantly greater when subjects watched live rather than recorded dance solos ([Jola and Grosbras, 2013](#)), and another TMS study found that the corticospinal excitability of a hand muscle was significantly greater when subjects watched live rather than recorded intransitive hand movements ([Prinsen and Alaerts, 2019](#)). These results indicate that MNS responses are often enhanced when actions are witnessed "in the here and now".

A related consideration is that almost all of the research on the MNS

has used two-dimensional (2D) videos as stimuli, but such stimuli do not allow onlookers to fully recover the precise kinematics of actions. To address this limitation, two recent fMRI studies showed subjects recorded actions in a stereoscopic manner so that three-dimensional (3D) nuances could be detected ([Jastorff et al., 2016](#); [Ferri et al., 2016](#)). Together, these studies found that compared with 2D actions, 3D actions evoked stronger responses in both parietal and frontal components of the MNS, with the greatest effects occurring in a ventral sector of the premotor cortex that not only displayed remarkable sensitivity to kinematic details (including accelerations), but is also known to receive vestibular input. As [Ferri et al. \(2016, p. 3607\)](#) point out, this last finding may help explain why "vestibular symptoms such as dizziness or nausea have been reported while viewing 3D movies...."

In the current context, it's also worth discussing the observation of actions performed by point-light agents—i.e., agents that are portrayed only by small lights placed on their major body parts, thereby capturing kinematics but not form. Despite having minimal explicit content, such stimuli convey a vivid impression of figures in motion, enabling observers to recognize not only a wide range of actions, such as walking, stretching, jumping, and throwing, but also many social characteristics, such as dominance, openness, vulnerability, and intent to deceive (for a review see [Blake and Shiffrar, 2007](#)). Interestingly, fMRI research indicates that both parietal and frontal components of the MNS respond to point-light actions ([Lestou et al., 2008](#)). Moreover, when subjects are asked to detect point-light walkers amid noisy backgrounds, not only does their performance improve as the amount of gray matter increases in their ventral premotor cortex ([Gilaie-Dotan et al., 2013](#)), but their performance is also disrupted by either repetitive TMS ([van Kemenade et al., 2012](#)) or structural damage ([Saygin, 2007](#)) to this region. In addition, a MEG study found that in overlapping observation-execution areas, human actions presented merely as point-light stimuli elicited the very same responses as human actions presented in full view ([Press et al., 2011](#)).

3. Factors involving the actor

3.1. Factor 7: similarity to humans: robots, animals, and other simple shapes

How human-like must an agent be in order to trigger MNS responses in a human observer? Do artificial agents, such as robots, evoke such responses, and if so, how do these responses compare with those evoked by humans? Also, what about non-human animals, such as dogs, and inanimate entities, such as geometric shapes?

Several studies have examined how the MNS reacts to robot actions vs. human actions, but the results are inconsistent (for a review see [Wykowska et al., 2016](#)). Two functional neuroimaging studies, one using PET ([Tai et al., 2004](#)) and the other fMRI ([Miura et al., 2010](#)), found significantly greater premotor responses to human than robot actions. However, while another fMRI study likewise found stronger premotor engagement for human than robot actions, this was only when the actions involved simple movements; in contrast, when the actions were complex and goal-oriented, equally strong responses to the two types of agents were found in parietal, premotor, and inferior frontal components of the MNS ([Gazzola et al., 2007a](#)). Similarly, a TMS study found equivalent increases of corticospinal excitability during the observation of human and robot actions ([Hogeveen and Obhi, 2012](#)), and an EEG study found equivalent mu rhythm (8–13 Hz) suppression—a putative MNS marker that is manifested over sensorimotor sites ([Fox et al., 2015](#); but see also [Hobson and Bishop, 2016](#), and [Coll et al., 2017](#))—when human and robot actions were viewed in a neutral context ([Oberman et al., 2007](#)). More surprisingly, an fMRI study found that the anterior intraparietal sulcus was engaged to a greater degree by the sight of a dancing robot than the sight of a dancing human, regardless of whether the kinematics were rigid or fluid ([Cross et al., 2012](#)). And another fMRI study found that both parietal and premotor

regions responded significantly more to a robot grasping things than to a human grasping things (Kupferberg et al., 2018). It's also notable that MNS responses to robot actions are sensitive to task demands (Chaminade et al., 2010) and to how closely the robot resembles a human (Saygin et al., 2012).

A few studies have compared MNS responses to actions performed by humans and other animals, but the results are, once again, rather mixed. First, in an fMRI study Buccino et al. (2004) showed subjects two sets of actions performed by a human, a monkey, and a dog. One set involved biting actions, which were similar across the three species, and the other set involved oral communicative actions, which were increasingly different across the three species—namely, a human talking, a monkey lip-smacking, and a dog barking. Relative to static images of the three agents, the biting actions evoked similar MNS responses in some parietal and frontal regions regardless of species, but the oral communicative actions did not; instead, while the sight of a person talking engaged the pIFG strongly and extensively, the sight of a monkey lip-smacking elicited a much weaker and smaller response in this region, and the sight of a dog barking did not activate it at all. According to the researchers, these results suggest that the MNS is recruited primarily by actions that belong to the observer's own motor repertoire (see also *Factor 13*).

More recently, two TMS studies explored this topic further. White et al. (2014) measured the corticospinal excitability of subjects' hand muscles while they watched grasping actions performed by a human, a rat, an elephant, and a robotic arm. The results were hard to interpret, however, because MNS responses only occurred for the rat and elephant actions. In contrast, Amoroso and Urgesi (2016) obtained more meaningful results when they examined how subjects who were either familiar or unfamiliar with dogs covertly reacted to videos of a human and a dog performing three kinds of actions—namely, greeting (by waving and tail-wagging, respectively), grasping (with the arm/hand and mouth, respectively), and offering to shake (with the arm/hand and foreleg/paw, respectively). They found that although the subjects who lacked significant experience with dogs only displayed increased corticospinal excitability of the flexor carpi radialis (FCR) muscle, which supports arm and wrist movements, when they watched the three human actions, the subjects who did have substantial experience with dogs displayed such mirror responses to both the human actions and the canine actions. This robust familiarity effect suggests that the MNS can be "tuned" to resonate to non-human actions. Notably, however, because the mirror responses to all three canine actions involved the arm/wrist-related FCR muscle, but two of those actions were performed with non-arm/wrist effectors—specifically, the dog's tail (an effector that humans lack) and the dog's mouth—the researchers propose that the motor resonance did not represent the actions at the level of movement kinematics, but rather at the level of intentional goals. (This is similar to Senna et al.'s (2014) finding that, as described in connection with *Factor 2*, the sight of a grasping action performed with a foot increased the corticospinal excitability of a relevant hand muscle in the observer, suggesting motor resonance at the goal level).

Finally, there is some evidence that the MNS is even sensitive to the movements of very simple inanimate entities, like geometric shapes. For instance, an fMRI study revealed that parietal, inferior frontal, and premotor components of the MNS were activated significantly more when clouds of moving dots complied with the two-thirds power law (described above in the subsection on *Factor 3*) than when they didn't, thereby supporting the notion that "common codes and/or representations constrain both the interpretation of visual and kinesthetic inputs and the generation of actual movements" (Dayan et al., 2007, p. 20585). In addition, another fMRI study found that both parietal and premotor components of the MNS responded to the movements of not only human hands but also yellow spheres (Engel et al., 2008a). Notably, the responses to spheres were weaker than those to hands, and a subsequent study showed that their magnitude was greater during a simulation task than a color judgment task (Engel et al., 2008a). Nonetheless, the studies just mentioned clearly show that the MNS is in fact engaged by the visual

motion patterns of rudimentary objects. Furthermore, in yet another fMRI study Ramsey and Hamilton (2010) found that when subjects watched a simple shape "take" one object rather than another, there was significant activation in the aIPS, which is known to distinguish between different target objects of transitive actions (see *Factor 2*). More generally, other research suggests that the motor system contributes to the perception and understanding of physical events triggered by natural or mechanical forces (Press and Cook, 2015). In this vein, Schubotz (2007) argues that our premotor cortex helps us predict events that we can't imitate, like the rolling of ocean waves, and that it does so by extending its rough somatotopic organization to these phenomena. Supporting this view, Schubotz and von Cramon (2003) found that serial prediction tasks involving rhythm tap mouth-related premotor areas, those involving objects tap hand-related premotor areas, and those involving spatial locations tap arm-related premotor areas.

3.2. *Factor 8: number of agents*

Almost all MNS research has focused on situations in which observers see the actions of just one agent. Two recent studies, however, demonstrated that dual-agent actions engage the MNS more than single-agent ones (Cracco et al., 2016, 2019).

First, Cracco et al. (2016) conducted a TMS study in which subjects passively viewed two right hands side by side, either one or both of which performed either an index finger abduction or a little finger abduction. Consistent with previous work, when the subjects saw just one hand execute each action, there was significantly greater corticospinal excitability of the relevant muscle than the irrelevant muscle. Shedding new light on the nature of the MNS, however, the investigators also discovered that when the subjects saw both hands execute identical actions, their mirror responses were even greater. This suggests that the well-known social-psychological phenomenon of group contagion—i.e., people's tendency to copy the behaviors of groups more than individuals—may arise because, as Cracco et al. (2016, p. 1422) put it, actions executed by groups simply "resonate louder".

In a subsequent fMRI study, Cracco et al. (2019) modified their design to examine how the MNS reacts to the sight of two agents simultaneously performing different actions. While monitoring for brief video flickers, subjects watched two right hands side by side, either one or both of which performed complex sign language gestures that were unrecognizable. Both parietal and premotor components of the MNS were activated more strongly when the two hands performed different gestures than when just one hand performed a single gesture. Moreover, greater activation was found in the anterior cingulate cortex, which is associated with motor and cognitive conflict, when the two hands performed different gestures than when they performed the same gesture. Finally, multivariate pattern analyses revealed that all of the observed gestures could be decoded from activity in the pertinent parietal and premotor regions. Taken together, these results not only provide further evidence that the MNS is modulated by the number of agents, but also indicate that it can represent different observed actions at the same time (for a broader perspective on the neuroscience of perceiving interacting agents, see Quadflieg and Koldewyn, 2017).

4. Factors involving the observer

4.1. *Factor 9: awareness*

As indicated in Fig. 2, the MNS is modulated by more observer-related factors than any other type. We will begin working our way through this long list by first considering whether the MNS requires conscious awareness of perceived actions in order to be engaged. So far only three studies have directly addressed this issue, but their results, which derive from different brain mapping methods, are largely consistent. In general, they suggest that although the MNS responds more strongly to consciously than unconsciously perceived actions, it is

also sensitive to the latter.

In a TMS study [Mattiassi et al. \(2014\)](#) recorded the corticospinal excitability of subjects' finger muscles while they passively observed static images of either still hands, finger movements related to the subjects' recorded muscles (henceforth "related actions"), or finger movements unrelated to the subjects' recorded muscles (henceforth "unrelated actions"). These consciously perceived probe stimuli were preceded by unconsciously perceived prime stimuli that consisted of the same static images and that were prevented from reaching awareness by means of both forward and backward masking. There were three main results. Focusing first on the conditions when the probe stimuli were related actions, corticospinal excitability was facilitated when the prime stimuli were either still hands or related actions. This shows that motor resonance occurred for the consciously perceived hand actions regardless of whether the unconsciously perceived primes were still hands or identical hand actions. Second, shifting to the conditions when the probe stimuli were still hands, corticospinal excitability was *not* facilitated when the primes were either related or unrelated hand actions. This suggests that unconsciously perceived hand actions do *not* evoke MNS responses. Third, returning to the conditions when the probe stimuli were related actions, corticospinal excitability turned out to be significantly weaker when the primes were unrelated actions than when they were either related actions or still hands. Thus, even though unconsciously perceived hand actions do not themselves elicit mirror responses in the primary motor cortex, they do reduce such responses to consciously perceived incongruent hand actions, perhaps via higher-order cortical regions that register action conflicts. These findings imply that the MNS does have some degree of sensitivity to subliminally processed actions.

Moving on to other work, in two "companion" EEG and fMRI studies, [Simon and Mukamel \(2016, 2017\)](#) simultaneously presented subjects with action videos through one eye and dynamic noise through the other eye. This technique rendered the action videos inaccessible to conscious awareness. In a separate condition the same videos were shown to subjects in a fully conscious fashion. The EEG study revealed that over sensorimotor sites mu suppression (8–13 Hz)—which, as mentioned earlier, is often regarded as an MNS marker—was significantly greater during conscious than unconscious action perception. Moreover, beta suppression (15–25 Hz)—which is also a putative MNS marker—displayed a similar profile; however, the oscillatory power of this rhythm was even suppressed during unconscious action perception relative to baseline. According to the authors, "the intensity of mirror system responses during action observation decreases with the observers' perception level yet remains significant during observation of invisible actions" ([Simon and Mukamel, 2016](#), p. 1). The aim of the subsequent fMRI study was to examine the neural correlates of perception level more closely, using a similar experimental design. The results suggested that within the MNS both the primary motor cortex and the dorsal premotor cortex are sensitive to perception level, and it's notable that the former region was the focus of [Mattiassi et al.'s \(2014\)](#) TMS study, and the latter region is one source of the mu rhythm ([Arnstein et al., 2011](#)). In contrast, the ventral premotor cortex, pIFG, and superior parietal lobule were found to be insensitive to perception level.

4.2. Factor 10: attention

Attention is, of course, closely related to awareness, and many studies have found that it too modulates the MNS. In general, when people's attention is directed toward the mechanical aspects of observed actions, MNS responses are enhanced, and when their attention is diverted away from those aspects, MNS responses are reduced. As we will see, however, when social situations call for complementary rather than imitative actions, the ways in which attention influences MNS activity are more complex.

Evidence for attentional augmentation of MNS activity comes from several sources. For instance, two fMRI studies found greater responses

in inferior parietal and ventral premotor regions when subjects made "how" judgments about observed actions than when they made "why" judgments about the very same actions ([Hesse et al., 2008](#); [Spunt et al., 2011](#)). And another fMRI study likewise found greater responses in inferior parietal and ventral premotor regions when subjects determined whether the movements of inanimate spheres could be simulated with human hands than when they determined whether the colors of those spheres changed ([Engel et al., 2008b](#); see also *Factor 7*). In addition, an EEG study found greater mu suppression when subjects counted how many times they saw a certain type of grip in a series of grasping actions than when they counted how many times they saw a certain type of color in the same series of grasping actions ([Schuch et al., 2010](#)). And a MEG study found greater beta suppression when subjects watched sequences of finger movements with the intent to later imitate them than when they just passively watched the same sequences of finger movements ([Muthukumaraswamy and Singh, 2008](#)). Furthermore, another MEG study found that although both attended and unattended point-light walkers evoked high gamma activity over the left occipitoparietal territory at 80 ms post-stimulus, only attended ones also evoked such activity over the right parietal cortex at 120 ms post-stimulus ([Pavlova et al., 2006](#)). Finally, two TMS studies found that the corticospinal excitability of relevant finger muscles was facilitated more when subjects observed actions with a fixed point-of-gaze ([D'Innocenzo et al., 2017](#)) and when they focused on how objects were being affected ([Wright et al., 2018](#)) than when they saw the same actions during free viewing.

Importantly, there's also evidence that when attention is shifted away from the physical features of observed actions, MNS responses are often still present, but are weaker and less precise. For example, a TMS study found that, relative to when subjects saw grasping actions at the center of their visual field, their motor resonance was rough and inaccurate when they saw the same actions in their visual periphery ([Leonetti et al., 2015](#)). Another TMS study found that, relative to when subjects passively watched an agent lift objects of different weights, their motor resonance was reduced when they simultaneously performed a secondary visual task but not when they simultaneously performed a secondary auditory task—results which suggest that motor resonance does require some selective visual attention but is otherwise nearly automatic ([Saucedo Marquez et al., 2011](#)). Similarly, yet another TMS study found that, relative to when subjects passively watched an agent execute rhythmic wrist movements, their mirror responses retained muscle specificity but lost kinematic precision when they simultaneously performed separate visual tasks that required focusing on an LED light on one of the agent's fingers ([Puglisi et al., 2017](#)). It's also notable that an fMRI study found that when subjects observed grasping actions while simultaneously performing high-load vs. low-load secondary visual tasks, activity in the bilateral inferior parietal cortices and the right pIFG was consistently high, perhaps reflecting automatic mirror responses, but activity in the left pIFG was reduced, perhaps reflecting a role for this region in filtering out task-irrelevant actions ([Chong et al., 2008](#)).

As mentioned above, the attentional modulation of motor resonance appears to be more complicated in social situations that involve complementary actions, which is to say, in situations where "two or more individuals coordinate and mutually complete their incongruent actions, rather than performing imitative behaviors" ([Sartori and Betti, 2015](#)). This was recently demonstrated in a study by [Betti et al. \(2017\)](#) that combined eye-tracking and TMS. There were two conditions, described by [Betti et al. \(2017, p. 4\)](#) as follows:

- *No-Request condition*: The actor grasps a sugar spoon lying on the table in front of her, pours sugar into a mug next to her, and then moves the spoon with some sugar left back to its initial position.
- *Request condition*: The actor grasps a sugar spoon lying on the table in front of her, pours sugar into a mug next to her, and then stretches out the spoon with some sugar left toward a mug out of her reach but

strategically placed near the observer, thus requiring the observer's intervention to lift the mug and complete the complementary action.

At the end of each condition, a red dot was flashed at either of two locations to momentarily capture the observer's attention: *left side*, over the actor's hand moving back to its initial position in the No-Request condition, and at the same position in the Request condition; and *right side*, over the out-of-reach mug on the right side of the screen near the observer in both conditions. There were two main results. First, attentional capture by the dot modulated mirror responses to the actor's use of the spoon. Specifically, the corticospinal excitability of a muscle for precision grip was not only greater in the No-Request condition when the dot was on the left side (superimposed on the actor's hand) than when it was on the right side (superimposed on the mug near the observer), but was also greater when the dot was on the left side in the No-Request condition (again, superimposed on the actor's hand) than when it was on the left side in the Request condition (where the actor's hand was not located). Second, and more strikingly, attentional capture by the dot did *not* affect the observers' implicit preparation of a complementary action—namely, grasping and lifting the nearby mug—in the Request condition. In particular, the corticospinal excitability of a muscle for whole-hand grip was significantly higher in the Request condition than the No-Request condition not only when the dot was on the right side (superimposed on the mug near the observer), but also when it was on the left side (where the actor's hand was not located). This last finding has serious implications, since it shows that when social situations call for complementary rather than imitative actions, "motor preparation is impervious to attentional interference" (Betti et al., 2017, p. 17).

4.3. Factor 11: perspective

Another influential factor is the perspective or point of view from which an observer perceives an action. This has been explored in both monkeys and humans, but even though the vast majority of studies do suggest that perspective modulates the MNS, the specific results are not completely consistent.

Monkey studies have focused on the premotor area F5 and the anterior intraparietal area AIP. Caggiano et al. (2011) recorded single-cell activity in F5 while two monkeys watched grasping actions from three perspectives: lateral, when the observer sees the agent from the right or left; allocentric, when the observer sees the agent facing him or her; and egocentric, when the observer sees the agent's hand movements from a 1st-person point of view. Some cells had equally strong responses across all three perspectives, but others responded preferentially to just one. Interestingly, although the percentages of perspective-specific cells did not differ significantly, in a follow-up study Caggiano et al. (2015) analyzed the local field potential in F5 and found that there was greater low-frequency power for the egocentric than the allocentric perspective (the lateral perspective was not included). Moreover, Maranesi et al. (2017) found that at both the single-cell level and the population level in F5, there were greater responses to grasping actions seen from an egocentric than a lateral perspective—a difference that was most pronounced when the actions were viewed in the on-looker's peripersonal space. Turning to AIP, Lanzilotto et al. (2020) measured single-cell firing rates while two monkeys observed seven manipulative actions (drag, drop, grasp, push, roll, rotate, and squeeze) in four visual formats involving the combination of two perspectives (lateral and allocentric) and two body postures (standing and sitting). Some populations of neurons had response properties that enabled highly accurate decoding of not only particular actions but also particular perspectives and postures (see also the computational model by Oh et al., 2019). Finally, a recent fMRI study by Fiave and Nelissen (2021) searched for but failed to find distinctive activation patterns for lateral, allocentric, and egocentric perspectives in both F5 and AIP. The lack of a clear preference for the egocentric viewpoint in F5 contrasts with

Caggiano et al.'s (2015) and Maranesi et al.'s (2017) results, but it could be due to the lower spatial resolution of fMRI.

With regard to human research, three studies—one using fMRI (Héту et al., 2011) and two using TMS (Burgess et al., 2013; Fitzgibbon et al., 2014)—did not find significant effects of perspective on the MNS; however, drawing on diverse brain mapping methods, many other studies have identified such effects, with the details varying across investigations. In general, the most common result is that, consistent with the monkey data reported by Caggiano et al. (2015) and Maranesi et al. (2017), MNS responses are greater for actions observed from an egocentric than an allocentric or lateral perspective. This has been found in multiple studies using fMRI (Jackson et al., 2006; Pilgramm et al., 2010; Oosterhof et al., 2012; Ge et al., 2018), EEG (Drew et al., 2015; Angelini et al., 2018), and TMS (Maeda et al., 2002). Two other studies—one using MEG (Kilner et al., 2006) and the other EEG (Frenkel-Toledo et al., 2013)—obtained what initially appear to be contrary results, namely greater MNS responses for actions observed from an allocentric than an egocentric perspective; however, in both studies the egocentric perspective was defined as seeing the whole agent from behind, so the relatively enhanced mirror activity that emerged for the allocentric perspective, which was defined as seeing the whole agent from the front, may have reflected the agent's greater social salience in that condition.

Another intriguing yet complex set of findings involves interactions between the observer's perspective (egocentric vs. allocentric) and the hand that the agent moves (right or left). When actions are perceived from a 1st-person egocentric perspective—i.e., as movements of the agent's right or left hand directed away from the observer, as if the observer were performing the actions himself or herself—they can be mapped directly onto the observer's own motor system without any spatial transformation, allowing anatomically congruent simulations of the perceived movements. In such cases, when right-handed subjects see right-hand actions, they typically display predominantly left-lateralized neural responses in parietal and frontal nodes of the MNS (Hesse et al., 2008; Shmuelof and Zohary, 2008; Alaerts et al., 2009; Cabinio et al., 2010; Vingerhoets et al., 2012). However, when they see left-hand actions, their MNS activity is more variable across studies, often involving predominantly right-lateralized neural responses (Hesse et al., 2008; Shmuelof and Zohary, 2008; Vingerhoets et al., 2012), but sometimes involving less clear-cut hemispheric asymmetries (Alaerts et al., 2009; Cabinio et al., 2010). Studies of how left-handed subjects perceive actions from an egocentric perspective are discussed below in the subsection on Factor 12.

When actions are perceived from a 3rd-person allocentric perspective—i.e., as movements of the agent's right or left hand directed toward the observer, as if the agent were facing the observer—they must be spatially transformed in order to be copied onto the observer's own motor system. In such cases, several fMRI studies have found that for right-handed subjects the MNS, especially in the parietal cortex, tends to avoid spatial transformation and instead represent the observed actions in a spatially compatible (a.k.a. specular or mirror-like) manner, as manifested by predominantly right-lateralized neural responses to observed right-hand actions and predominantly left-lateralized neural responses to observed left-hand actions (Hesse et al., 2008; Shmuelof and Zohary, 2008; Vingerhoets et al., 2012). These findings fit nicely with many behavioral studies of overt imitation which have shown that when subjects explicitly mimic the actions of a model seen from an allocentric perspective, they tend to sacrifice anatomical congruence for spatial compatibility (e.g., Bekkering et al., 2000; Wohlschläger and Bekkering, 2002; Avikainen et al., 2003; Chiavarino et al., 2007). A TMS study also found that right-handed subjects displayed spatially compatible primary motor activity when they saw grasping actions from an allocentric perspective; however, this was restricted to left-lateralized neural responses to left-hand actions (Alaerts et al., 2009). Subsequently, another TMS study found that when right-handed subjects saw grasping actions from an allocentric perspective, primary

motor activity was again limited to left-lateralized neural responses, but this time such responses occurred for both left-hand and right-hand actions—results which are inconsistent with both spatial compatibility and anatomical congruence and suggest instead that the observer's own hand preference may be the main factor driving primary motor responses to allocentrically perceived grasping actions, regardless of which hand the agent uses (Sartori et al., 2013). It remains unclear, though, how these TMS findings can be reconciled with the fMRI findings described above. Studies of how left-handed subjects perceive actions from an allocentric perspective are discussed below in the subsection on *Factor 12*.

Finally, MNS responses are also influenced by how closely and naturally observers see actions. In a TMS study Bunday et al. (2016) found that the corticospinal excitability of pertinent muscles was facilitated when subjects saw a disembodied hand execute either a precision grip or a whole-hand grasp, but not when they saw the entire person perform the same actions. Furthermore, in an EEG study Heimann et al. (2014) found that when subjects saw hand actions which were presented as videos that replicated the natural vision of a person walking toward the agent, there was greater mu suppression than when they saw the same hand actions filmed from a fixed position.

4.4. Factor 12: handedness

Virtually all MNS research has been restricted to right-handed subjects. However, a few studies have directly compared right-handers with left-handers and found significant group differences.

Two fMRI studies measured activation patterns in the MNS while both right-handers and left-handers not only executed right-hand and left-hand actions, but also observed, from an egocentric perspective, such actions being performed by another agent (Willems and Hagoort, 2009; Cabinio et al., 2010). Although the two studies analyzed the data in different ways, they both found that in parietal and premotor components of the MNS, and during both execution and observation, right-handers displayed more lateralized activation than left-handers for both right-hand and left-hand actions.

In the second study, however, Cabinio et al. (2010) also divided the group of left-handers into two subgroups—"mildly" left-handed ($N = 9$) and "strongly" left-handed ($N = 11$)—in order to compare their results separately with those of the right-handers. This revealed three different patterns of lateralization:

- The right-handers were always relatively left-lateralized, but more so for right-hand than left-hand actions.
- The mild left-handers were mostly bilateral, with a slight left-lateralization for right-hand actions and a slight right-lateralization for left-hand actions.
- The strong left-handers were always relatively right-lateralized, but more so for left-hand than right-hand actions.

The finding that right-handers had left-lateralized neural responses even to left-hand actions conflicts with other fMRI studies which found that, as noted above in connection with *Factor 11*, when right-handers see left-hand actions from an egocentric perspective, they exhibit more right-lateralized MNS activation, especially in the parietal cortex (Hesse et al., 2008; Shmuelof and Zohary, 2008; Vingerhoets et al., 2012). However, the discovery that the right-handers and the strong left-handers displayed diametrically opposed lateralization profiles suggests that, as Cabinio et al. (2010, p. 319) put it, "for complex hand tasks the observer's handedness is what constraints the motor command both in execution and in motor resonance, and the identity of the observed hand is not what determines the lateralization of brain activation, although it may have some influence on the degree of lateralization of responses".

Moreover, despite the fact that the subjects in Cabinio et al.'s (2010) fMRI study observed actions from an egocentric perspective, the results

are consistent with those from Sartori et al.'s (2013) TMS study, in which both right-handers and left-handers observed both right-hand and left-hand actions from an allocentric perspective. We noted in the previous subsection that the right-handers in this study displayed left-lateralized primary motor responses to both types of actions. What's new here is that the left-handers displayed the converse hemispheric asymmetry—namely, right-lateralized primary motor responses to both types of actions. Together with Cabinio et al.'s (2010) fMRI findings, these TMS findings support the view that motor resonance is strongly influenced by the observer's own hand preference. According to Sartori et al. (2013, p. 6), the data point to "a sophisticated mechanism which converts another person's pattern of movement into optimal motor commands in an observer".

4.5. Factor 13: action familiarity and expertise

One of the most extensively studied factors modulating the MNS is whether or not—and if so, how well—observers can themselves perform the actions that they see. Many investigations have found that the more experience and skill observers have with a particular type of action, the more strongly their MNS is engaged when they see someone else produce it. However, a seemingly contradictory line of work has shown that the MNS is sometimes heavily recruited during the perception of actions that *cannot* be mapped directly onto the viewer's own motor repertoire. These points are elaborated below.

First of all, several fMRI studies suggest that as action familiarity increases, so does MNS engagement. For instance, Plata Bello et al. (2013) presented subjects with finger opposing movements involving the thumb and the other four fingers of a hand, and found that in inferior parietal and premotor components of the MNS, the degree of activation progressively increased as the movements became more familiar—i.e., as they shifted from the most unnatural and uncommon type of finger opposition (thumb against little finger) to the most natural and common type (thumb against index finger). Similarly, Gardner et al. (2015) showed subjects whole-body dance movements that varied in terms of their purely visual familiarity, and found that as familiarity increased, so did the amount of activation in inferior parietal and inferior frontal regions. Furthermore, Avanzino et al. (2015) conducted a TMS study in which subjects saw sequences of finger movements at different tempos, and found that the corticospinal excitability of a pertinent muscle was greatest when the tempo of the perceived movements matched the on-lookers' own preferred tempo for freely tapping out a rhythm.

In the same vein, there's also considerable evidence that as action expertise increases, so does MNS engagement. To be sure, people who have substantial visual experience with a particular domain of action, but who lack the skills necessary to perform such movements, nonetheless exhibit stronger MNS responses than complete novices during observation (Aglioti et al., 2008; Cross et al., 2009; Jola et al., 2012). However, both the magnitude and the precision of MNS responses are markedly greater in behavioral experts than in mere perceptual experts, as shown by studies involving a variety of action domains, such as dance (Calvo-Merino et al., 2005, 2006), basketball (Aglioti et al., 2008), and piano (Candidi et al., 2014). More generally, a number of other studies have used diverse brain mapping methods to demonstrate that—again, in multiple action domains—as one's ability to perform intricate actions improves with practice, so does one's MNS sensitivity to the sight of such actions, presumably reflecting the capacity for increasingly accurate motor simulations (dance: Cross et al., 2006; Pilgramm et al., 2010; Amoroso et al., 2014; Kirsch and Cross, 2015; Di Nota et al., 2017; sports: Balser et al., 2014; Makris and Urgesi, 2015; Özkan et al., 2010; Pedullà et al., 2020; piano: Haslinger et al., 2005).

The situation is far from straightforward, however, because a few other investigations have shown that the MNS is sometimes recruited more heavily by actions that are *not* in the observer's own motor repertoire than by actions that are in it. For example, in an fMRI study Cross et al. (2012) found that, to their surprise, inferior parietal and

premotor components of the MNS were engaged more robustly during the observation of rigid than fluid dance movements, regardless of whether the agent was a robot or a human. And in another fMRI study Liew et al. (2013) found that the inferior parietal node of the MNS responded more to actions performed by the residual limb of an amputee than to actions performed by a normal hand. Furthermore, in a recent TMS study Hilt et al. (2020) obtained results which suggest that during the observation of complex multijoint actions, the primary motor cortex highlights the differences, rather than the similarities, between the perceived movements and the onlooker's own motor templates. In this context, it's also notable that several fMRI and TMS studies have found that frontal regions involved in speech production make their greatest contribution to speech perception when the sensory signals are hard to process—e.g., when the input contains non-native phonemes (Wilson and Iacoboni, 2006; Callan et al., 2003, 2004; Schmitz et al., 2019), when the input has been degraded (Shahin et al., 2009), when the input is embedded in background noise (D'Ausilio et al., 2012), and when the talker's idiosyncratic articulatory patterns differ substantially from the listener's (Bartoli et al., 2015).

Finally, in two TMS studies Fiorio et al. (2010a, 2010b) showed that during the observation of unusual pathological actions, the viewer's motor system is differentially modulated in complex ways that depend on both visual familiarity and the ability to recognize suboptimal movement kinematics. In the first study, two groups of subjects—healthy individuals and patients with a task-specific form of dystonia, namely writer's cramp—watched videos of both normal and dystonic handwriting actions while TMS was applied to hand and forearm sites in the primary motor cortex (Fiorio et al., 2010b). The results revealed opposite patterns of corticospinal excitability of the hand-related FDI muscle in the two groups, such that the healthy subjects displayed greater facilitation in response to the dystonic than the normal actions, whereas the patients displayed greater facilitation in response to the normal than the dystonic actions. This study fits with those mentioned above which found that the MNS responds more strongly to actions that are *not* in the observer's own motor repertoire than to those that are in it. In the second study, however, the researchers went on to show that this phenomenon is sensitive to the observer's knowledge about the perceived atypical movements (Fiorio et al., 2010a). Once again, two groups of subjects watched videos of both normal and dystonic handwriting actions while TMS was delivered, but while one group consisted of naive healthy individuals, the other consisted of clinicians who regularly treat patients with movement disorders like dystonia. As in the first study, the naive subjects exhibited greater corticospinal excitability of the FDI muscle during the observation of dystonic than normal movements. However, the clinicians exhibited similar levels of FDI facilitation in the two conditions. According to Fiorio et al. (2010a, p. 695), these results suggest that "observing something unusual and unknown, such as pathological actions to naive eyes, induces corticospinal hyper-activation, whereas acquired visual expertise prevents this hyper-activation".

4.6. Factor 14: learned associations

Another factor that influences the MNS is visuomotor training. But while some studies suggest that such training can temporarily reverse or abolish typical mirror responses to observed actions, other studies suggest instead that arbitrary rule-based responses occur later than typical mirror responses, are mediated by prefrontal regions subserving cognitive control, and are closely related to so-called complementary actions.

In a series of experiments, Catmur et al. (2007, 2008, 2011) and Cavallo et al. (2014) showed that the MNS can be configured to generate new visuomotor mappings. For instance, in a TMS study Catmur et al. (2007) taught subjects to overtly respond to the sight of an index finger movement by producing a little finger movement, and to overtly respond to the sight of a little finger movement by producing an index finger movement. After the training was complete, they found that when

the subjects saw an index finger movement, the corticospinal excitability of their own little finger increased, and when they saw a little finger movement, the corticospinal excitability of their own index finger increased. Moreover, in a subsequent TMS study that used a similar training protocol, Cavallo et al. (2014) found that such counter-mirror responses occurred just as fast as typical mirror responses, beginning roughly 200 ms after the visual detection of finger movement—a striking result which raises the possibility that visuomotor training can overwrite normal patterns of motor resonance. A limitation of this study, however, is that the directions of the observed movements were always the same from the subjects' perspective, with the index finger always moving downward and the little finger always moving upward. Hence, as the authors acknowledge and as Ubaldi et al. (2015) emphasize, the subjects could have learned the new visuomotor mappings by using a purely spatial strategy: "When anything moves downward, move my little finger upward, and when anything moves upward, move my index finger downward".

Even more importantly, other experiments have yielded different results which point to a more refined view of how learned associations influence the MNS. In two TMS studies Barchiesi and Cattaneo (2012) and Ubaldi et al. (2015) found that after subjects had been trained to link particular types of observed movements with different types of executed movements, the observed movements triggered a biphasic pattern of corticospinal excitability with the following characteristics. The first phase involved an early stimulus-driven mirror response (beginning 150–200 ms post-onset) that covertly replicated the observed movements, was independent of their spatial orientation, and was mediated by the dorsal visuomotor pathway. In contrast, the second phase involved a later rule-based counter-mirror response (beginning 300–320 ms post-onset) that relied on the dorsolateral prefrontal cortex, which presumably activated the motor cortex in a top-down manner that was guided by the arbitrary stimulus-response mapping. These findings indicate that normal mirror responses to particular types of perceived actions are not necessarily eliminated when subjects are trained to associate those actions with incompatible ones. Instead, normal mirror responses appear to be triggered more rapidly and automatically than counter-mirror responses, and the latter are only selected for overt execution after the prefrontal cognitive control system has intervened (see also Naish and Obhi, 2015).

Nonetheless, it's worth noting that the capacity for counter-mirror responses is highly adaptive, because in many real-life social situations, especially those involving complementary actions between two people, imitative behaviors must be suppressed in favor of alternative ones (for a review see Sartori and Betti, 2015; see also the discussion of Betti et al.'s (2017) TMS study in the subsection on *Factor 10*, as well as the subsection on *Factor 22*). Even in these situations, however, covert mirror responses are still useful, since they facilitate each person's ability to track the other's actions, thereby enabling the selection and overt execution of appropriate counter-mirror responses, as when two members of a dance pair, or two partners in a construction project, take cues from each other to coordinate their separate movements.

4.7. Factor 15: expectations

The literature on how expectations modulate the MNS is quite large, and for this reason only some of the most prominent findings are discussed here.

A good place to start is with studies which show that when an observer knows that an agent is about to perform an action, his or her motor system becomes excited well before the agent begins to move, suggesting anticipatory motor simulation. For instance, in an EEG study Kilner et al. (2004) found that the mere sight of a color cue indicating that an agent was about to grasp an object was sufficient to cause the observer's own motor system to enter the same state of readiness it would have if he or she were about to execute the action. More recently, Maranesi et al. (2014) recorded the spiking of mirror neurons in

premotor area F5 while monkeys watched an agent either produce ("go") or withhold ("no-go") a grasping action. Crucially, an auditory cue allowed the monkeys to predict which choice the agent would make on each trial. On the "go" trials, although some mirror neurons discharged only after the onset of the agent's movement, others discharged nearly half a second sooner, providing further evidence for anticipatory motor simulation.

Turning to the observation of ongoing rather than imminent grasping actions, several TMS studies have explored the potential influence of expectations on the MNS, but the results are inconsistent. In one frequently cited study, Gangitano et al. (2004) examined how the corticospinal excitability of subjects' finger muscles was affected by the sight of an unnatural reach-to-grasp movement in which the agent's fingers first opened in the normal manner but then suddenly closed before opening again just before contacting the target object. The initial, normal phase of this action triggered motor resonance, but then the unpredicted finger closure abolished it. According to the researchers, the resonant motor plan was fully "loaded" in the observers' MNS when the action began, but then the deviation from that plan caused it to be "discarded" without replacement (Gangitano et al., 2004, p. 2200). Thus, this study supports the view that expectations influence motor resonance during the observation of ongoing grasping actions.

Two other TMS studies provide further evidence for this view. Craighero et al. (2014) found that when the kinematics of perceived reaching and grasping actions were kept constant but the target objects varied, the corticospinal excitability of the onlookers' hand muscles was only facilitated when the object's shape afforded an easy grip with the given hand configuration—specifically, when the object had flat sides rather than pointy ones. Relatedly, McCabe et al. (2015) presented subjects with videos of an agent reaching toward and grasping either a small object or a large one. When the subjects did not know what the target object would be on each trial, the corticospinal excitability of their hand muscles was only modulated by the kinematics of the actions, but when they were given advance knowledge of the target objects, their corticospinal excitability was sensitive to both the kinematics and the goals (i.e., target objects) of the actions. In both of these studies, much like in Gangitano et al.'s (2004) study, the observers' motor resonance was influenced by their assumptions, and hence expectations, about what sorts of grasping actions are appropriate for what sorts of target objects.

Three other TMS studies, however, have challenged this view. First, as in Gangitano et al.'s (2004) study, Gueugneau et al. (2015) showed subjects unnatural reach-to-grasp actions, but in contrast to the former study, they found that when the observed kinematics diverged from the predicted kinematics, the subjects' corticospinal excitability was not abruptly abolished; instead, it continued to track the observed kinematics, suggesting stimulus-based rather than expectation-based guidance of motor resonance. Second, Janssen et al. (2015) gave subjects either valid or invalid cues about whether they would see either a precision grasp or a whole-hand grasp, and in keeping with Gueugneau et al.'s (2015) results, they found that on the mismatch trials, when the observed kinematics diverged from the predicted kinematics, the subjects' corticospinal excitability followed the former rather than the latter. Finally, Soriano et al. (2019) showed subjects reach-to-grasp actions directed at either large or small objects, and at three time points—20 %, 50 %, and 80 % of movement duration—they not only measured the subjects' corticospinal excitability but also asked them to predict the size of the target object. They found that the patterns of motor facilitation evoked by the two types of actions did not differ significantly until the 80 % time point; however, the subjects' behavioral judgments were already significantly above chance at the 20 % time point. These findings indicate that, as the authors put it, "even when no discrepancies are revealed and even when the course of the observed action is fully predictable," corticospinal excitability mimicks the observed movements (Soriano et al., 2019, p. 7, emphasis in original). Overall, it is apparent that these three studies conflict with Gangitano

et al.'s (2004), Craighero et al.'s (2014), and McCabe et al.'s (2015) studies insofar as they suggest that motor resonance during the observation of grasping actions is not significantly modulated by expectations. Although the reason for these different results is not clear, Gueugneau et al. (2015) point out that the videos in Gangitano et al.'s (2004) study were artificially slowed down and hence may have encouraged predictive processing.

Inconsistent results have also come from a number of TMS studies that sought to determine which factors influence the encoding of muscular force in subjects' primary motor cortex when they watch agents grasp and lift objects of different weights. On the one hand, some studies suggest that in such situations the amplitude of the observers' corticospinal excitability is shaped more by kinematic cues regarding the agent's effort than by other cues that are sometimes misleading. For instance, Alaerts et al. (2010) showed subjects videos in which an agent lifted four containers that varied in their weight (heavy or light) and filling level (high or low). For two of the containers these variables were congruent—i.e., (i) heavy/high-filling and (ii) light/low-filling—whereas for the other two they were incongruent—i.e., (iii) heavy/low-filling and (iv) light/high-filling. The agent's motion trajectory always reflected each object's actual weight, and the observers' muscle-specific motor facilitation reliably matched that kinematic information rather than the object-intrinsic information based on filling level. Further evidence for the predominant influence of kinematic cues was provided by Senot et al. (2011), who found that subjects' motor facilitation was not significantly different in the following two conditions involving the observation of lifting actions: "visible," when an agent lifted two transparent bottles filled with a large or small amount of sand; and "hidden," when an agent lifted two opaque bottles filled with a large or small amount of sand.

On the other hand, there's also evidence that in some circumstances expectations do influence motor resonance during the observation of lifting actions. For example, Alaerts et al. (2012) found that when subjects repeatedly saw an agent reach toward, grasp, and lift an object with a known weight, their motor resonance encoded grip force even during observation of the reach phase of the action. Such results indicate that, as the authors put it, "action simulation not only mirrors the present state of an observed action, but also sets up an internal model in the observer to predict the forthcoming action ahead of its realization" (Alaerts et al., 2012, p. 511). In addition, Senot et al. (2011) found that when subjects watched an agent lift two opaque bottles that had the same weight but different labels—namely, "heavy" and "light"—their motor resonance essentially disappeared. This result may reflect a general inhibition of corticospinal excitability triggered by the detection of mismatching semantic and kinematic cues. As Senot et al. (2011, 165) put it, "any time a conflict is present between explicit and implicit movement-related information, the observer's motor system stops its mirroring of the observed action". More recently, Rens et al. (2020) likewise found that when observers' expectations about the weights of lifted objects were violated, their mirror responses were reduced. According to these researchers, such suppression "might be useful to prevent motor resonance from encoding object weight based on an incorrect prediction" (Rens et al., 2020, p. 4008).

Notably, these findings from Senot et al. (2011) and Rens et al. (2020) resemble Gangitano et al.'s (2004) and Craighero et al.'s (2014) finding that motor resonance was eradicated when an observed grasping action deviated from what was expected. But they are at odds with Gueugneau et al.'s (2015) and Janssen et al.'s (2015) finding that when such deviations are detected, motor resonance is not abolished but rather continues to reflect the perceived kinematics. These contradictions, among others, suggest that expectations sometimes do, and sometimes don't, modulate the MNS during action observation (for other relevant studies see Obhi and Hogeveen, 2010; Buckingham et al., 2014; Ondobaka et al., 2015; Jacquet et al., 2016; see also the comments about expectations in the subsections on *Factors 7* and *13*).

Finally, some additional insights come from two TMS studies that

investigated MNS activity while subjects watched agents lift either heavy or light objects in either a genuine or a deceptive manner (Tidoni et al., 2013; Finisguerra et al., 2018). First, Tidoni et al. (2013) applied repetitive TMS to the pIFG and found that the subjects' ability to discriminate between genuine and deceptive actions was impaired (though their performance on a difficulty-matched spatial control task was unaffected). This discovery dovetails with other studies that have implicated the pIFG in the attribution of intention (see the subsection on Factor 2). Furthermore, in another experiment Tidoni et al. (2013) applied single-pulse TMS to the primary motor cortex and found that the corticospinal excitability of subjects' hand muscles was facilitated more during the observation of deceptive than genuine actions, regardless of object weight. It was not clear, however, whether this effect was due to low-level kinematic features or high-level intentional features of the perceived actions. To resolve this issue, Finisguerra et al. (2018) independently manipulated these two levels of representation and found that while kinematic alterations were associated with muscle-specific activity, intentional alterations were only associated with muscle-independent activity. These results suggest that both low-level kinematic features and high-level intentional features of observed actions are coded in the onlooker's primary motor cortex, but in different ways.

4.8. Factor 16: prior social interaction

This factor has only been examined in a single study, but it is nonetheless important enough to warrant inclusion in the current survey. Hogeveen and Obhi (2012) investigated whether naturalistic social interaction modulates the MNS during subsequent action observation. In the first part of the study, one group of subjects interacted with a researcher in a music rating task. The researcher played four fragments from Miles Davis's *Kind of Blue* for each subject and asked him or her to describe the fragments with three adjectives. Also, during this period the researcher touched his face approximately once every 10 s, and the degree to which each subject spontaneously mimicked this face-touching behavior was recorded. In the second part of the study, the subjects who had just participated in the social interaction, as well as another group of subjects who had not participated in it, took part in a TMS experiment. Specifically, the corticospinal excitability of relevant finger muscles was measured while the subjects observed ball-squeezing behaviors performed by either a human hand or a robot hand. The results revealed that the subjects who had just engaged in social interaction displayed greater motor resonance for the human than the robot actions, whereas the subjects who had not engaged in prior social interaction displayed similar motor resonance for both kinds of actions. Moreover, among the subjects who had engaged in social interaction, the magnitude of motor resonance for the human actions was greater for those subjects who tended to mimic the researcher's face-touching behavior than for those who did not. Hogeveen and Obhi (2012, p. 5988) concluded their paper as follows: "In summary, we have shown that social interaction enhances biological tuning of the motor resonance system, and our ... data suggest that mimicry during the social interaction increases this effect".

4.9. Factor 17: motivation, emotion, personality, and morality

One of the most intriguing aspects of the MNS is that, as many recent studies have shown, it is modulated by what may be called, in general terms, the "subjective value" that an action has for an observer. As noted in the Introduction, this is a broad topic that could in principle be broken down further into a number of separate factors that focus on different value-based influences. For the sake of simplicity, however, the current section lumps together a variety of them, including motivation, emotion, personality, and morality (see also Aziz-Zadeh et al., 2018).

Before delving into studies with humans, it's worth mentioning a study with macaque monkeys which found that some mirror neurons in

area F5 are influenced by the subjective value of the target objects of perceived grasping actions. In particular, Caggiano et al. (2012) discovered that some mirror neurons in this region respond strongly during the observation of grasping actions directed at high-value objects (i.e., preferred foods as well as non-food objects predictive of reward), whereas other mirror neurons in the same region respond strongly during the observation of grasping actions directed at low-value objects (i.e., non-food objects not associated with reward).

Turning now to human studies, let's begin with motivation. We have already seen that although the MNS is affected by attention (see Factor 10), it is still a fairly automatic process, being triggered by unconsciously perceived actions (see Factor 9) and persisting in the normal manner even after counter-mirror training (see Factor 14). It should therefore come as no surprise that people cannot easily increase or decrease their motor resonance to perceived actions through sheer willpower (Naish and Obhi, 2015). Nonetheless, there is some evidence that motivation—or, more precisely, the degree to which someone would like to perform an observed action—can influence MNS responses in an implicit manner. This was demonstrated in an fMRI study by Cheng et al. (2007). In two separate scanning sessions separated by 1.5 h, they showed two groups of subjects videos of an agent either grasping food or grasping other objects. Crucially, the subjects in group 1 had fasted before the first session but were allowed to eat before the second; in contrast, the subjects in group 2 had a meal before the first session. The researchers found that when the subjects in group 1 were hungry, several parietal and frontal components of the MNS were engaged more by the food-related than the non-food-related actions. Moreover, there was a positive correlation between these subjects' self-reported hunger ratings and their degree of pIFG recruitment, which fits with other research implicating this region in the representation of the ultimate purposes of actions (see Factor 2).

Shifting to emotion, it too significantly modulates the MNS. For instance, several fMRI and TMS studies have shown that MNS responses to observed actions are especially strong when the onlooker is in an emotional state characterized by both negative valence and high arousal (e.g., anger or disgust), induced by either the context of the observed action or prior exposure to affective stimuli (Enticott et al., 2011, 2012; Hill et al., 2013; Mazzola et al., 2013; Nogueira-Campos et al., 2016; Lagravinese et al., 2017). The reason why such an emotional state elevates MNS responses so much is not clear, but it may be the first step of a defensive reaction to a potentially threatening situation. Social emotions also influence the MNS, with the most common finding being that subjects with relatively high status and power display weaker MNS responses to observed actions than subjects with relatively low status and power (Hogeveen et al., 2014; Varnum et al., 2016; Farwaha and Obhi, 2019). For example, Farwaha and Obhi (2019) conducted a TMS study involving subjects who were classified according to their Instagram profiles as either "leaders" (i.e., individuals who had more Instagram users following them than they followed) or "followers" (i.e., individuals who had fewer Instagram users following them than they followed). The researchers first required each subject to explicitly acknowledge his or her online status as either a leader or a follower, and then they recorded each subject's muscle-specific corticospinal excitability in response to a video of an agent squeezing a ball. They found that the leaders had markedly less motor facilitation than the followers—a result which suggests that, on average, individuals with relatively high social rank may have less "interpersonal sensitivity" than individuals with relatively low social rank (Farwaha and Obhi, 2019, p. 5).

A few recent studies have shown that the MNS is also affected by the sorts of attitudes and traits that contribute to selfhood. Two TMS studies are particularly relevant here. In one of them, Obhi et al. (2011) demonstrated that motor resonance can be dynamically shifted over the course of mere seconds by priming subjects with words associated with either an interdependent type of self-construal (which defines the self in terms of relationships like father, daughter, coworker, etc.) or an independent type of self-construal (which defines the self in terms of intrinsic

qualities like friendly, compassionate, ambitious, etc.). Subjects' muscle-specific corticospinal excitability was recorded while they watched a video of an agent squeezing a ball, and on each trial an interdependent prime word (e.g., *together*), an independent prime word (e.g., *alone*), or no prime word was superimposed on the video. Relative to the no-prime condition, the interdependent-prime condition elicited greater mirror responses whereas the independent-prime condition elicited lower mirror responses, even when these three kinds of trials were only separated by seconds in mixed blocks. (This discovery is especially fascinating in light of [Henrich's \(2020\)](#) argument that ancestral human societies were predominantly collectivist rather than individualist.) In a closely related study, [Liuzza et al. \(2015\)](#) discovered that motor resonance is influenced by a combination of the personality traits of the observer and the morality of the action. They recorded subjects' muscle-specific corticospinal excitability in response to two grasping actions with identical kinematics but different moral connotations—namely, stealing a wallet and picking up a notepad. The results revealed that motor resonance was selectively suppressed for the immoral action in just those subjects who had high scores for "harm avoidance," a personality trait characterized by behavioral inhibition and a heritable tendency to be cautious and fearful.

Continuing with the theme of morality, its relevance to motor resonance was recently confirmed in another recent TMS study by [Craighero and Mele \(2018\)](#). The subjects were first familiarized with how the act of squeezing a ball had one of three effects on a cuff wrapped around an individual's upper arm: gradual inflation, causing discomfort; gradual deflation, causing relief; or no change. Next, while their muscle-specific corticospinal excitability was being recorded, they watched an agent repeatedly squeeze the ball in three different conditions: "Dislike," when the subjects believed the action was causing discomfort in someone else; "Like," when they believed the action was causing relief in someone else; and "Neutral," when they believed the action had no immediate consequences. The most striking result was that, completely apart from variability in the subjects' personality traits, their mirror responses were uniformly reduced in the Dislike condition. According to [Craighero and Mele \(2018\)](#), if an observer thinks that a particular action is bad because it hurts someone else, this ethical judgment has the potential to interfere, in a top-down manner, with the bottom-up process of generating a somatotopically and kinematically precise simulation of that action. In short, moral values influence motor resonance (see also [Aragón et al., 2014](#); [Etzel et al., 2016](#)).

5. Factors involving the relationship between actor and observer

5.1. Factor 18: postural congruence

Let's shift now to modulatory factors that involve actor-observer relationships. The first of these—namely, postural congruence—is closely related to *Factor 11*—namely, perspective—but it deals more directly and systematically with whether the bodily orientations of the actor and observer are aligned. As described below, although this factor has only been investigated in a few studies, the results suggest that it does influence the MNS, albeit through interactions with other variables involving the body and space.

In a TMS study by [Urgesi et al. \(2006\)](#), subjects watched an agent's right hand perform movements with either the index finger (controlled by the first dorsal interosseous [FDI]) or the little finger (controlled by the abductor digiti minimi [ADM]) while the corticospinal excitability of the corresponding muscles in their own right hand was recorded. Crucially, the agent's hand varied between palm-down and palm-up postures, and so did the subjects' hand, so that the agent's and subjects' hand postures were sometimes congruent and sometimes incongruent.

The researchers found that FDI excitability selectively increased during the observation of index finger movements, regardless of

whether the agent's hand was palm-down or palm-up, but only when the subjects' hand was palm-down. And in a symmetrically inverse manner, they also found that ADM excitability selectively increased during the observation of little finger movements, regardless of whether the agent's hand was palm-down or palm-up, but only when the subjects' hand was palm-up. Interestingly, in a separate experiment the researchers recorded the subjects' FDI and ADM excitability while they executed, rather than watched, index finger and little finger movements with both palm-down and palm-up postures, and the results were analogous to the observation conditions: greater FDI excitability when the hand was palm-down, and greater ADM excitability when it was palm-up, perhaps due to muscular or biomechanical aspects of these postures. Thus, within the context of this particular study, motor resonance was modulated by a complex interplay between muscle specificity and postural congruence between actor and observer—an interplay that, importantly, was shaped in part by constraints on action execution.

In another TMS study by [Alaerts et al. \(2009\)](#), subjects watched simple extension movements of an agent's right or left wrist while motor-evoked potentials from their own right extensor and flexor muscles were measured. The agent's hands were always palm-down, but the subjects' right hand varied between palm-down and palm-up postures. When the subjects saw right-hand extension movements, their extensor was facilitated more than their flexor, regardless of whether their hand was palm-down or palm-up. But when they saw left-hand extension movements, the results were partly different. Approximately half of the subjects displayed the same muscle-specific motor resonance that they displayed for right-hand extension movements, irrespective of postural congruence. In contrast, the other half displayed the following effect of postural congruence. When the hand orientations were congruent (i.e., agent's hand palm-down and subjects' hand palm-down), there was greater extensor than flexor facilitation, just like during the observation of right-hand extension movements. But when the hand orientations were incongruent (i.e., agent's hand palm-down and subjects' hand palm-up), there was greater flexor than extensor facilitation. The latter finding suggests that the subjects "resonated" with the observed movement according to its direction in extrinsic space. After all, when they saw an upward movement, excitability was highest in their own upward moving muscle, which was the flexor rather than the extensor, since their hand had a palm-up posture. It is not clear why this pattern was manifested only during the observation of left-hand extension movements and only by half of the subjects. Nonetheless, it bears emphasis that, as [Alaerts et al. \(2009, p. 420\)](#) point out, "direction specific (extrinsic) versus muscle specific (intrinsic) mapping can only be disentangled with incongruent posture conditions".

5.2. Factor 19: racial, ethnic, and abstract social group congruence

Another relational factor that modulates the MNS is whether the actor and observer belong to the same social group, as defined by racial, ethnic, or other criteria. The most common finding is that the MNS responds more strongly during the observation of actions performed by in-group than out-group members (for different results see [Déry and Théoret, 2007](#); [Losin et al., 2012](#)). For instance, a TMS study found greater FDI excitability when subjects saw hand actions performed by people whose race/ethnicity was the same as, rather than different from, their own ([Molnar-Szakacs et al., 2007](#)). Similarly, two fMRI studies found that the inferior parietal component of the MNS was engaged more strongly when subjects saw hand actions performed by people whose race/ethnicity was the same as, rather than different from, their own ([Liew et al., 2011](#)).

There are also some hints that the MNS is sensitive to aspects of social group categorization that transcend the superficial features of race/ethnicity and reflect abstract criteria instead. In an fMRI study [Molenberghs et al. \(2013\)](#) divided a group of Caucasian men into two teams that competed to determine which was faster at reaching toward and pressing a button. When the subjects were subsequently shown videos of

the reach-to-press movements performed by members of each team and had to judge their speed while being scanned with fMRI, those subjects whose judgments revealed an own-team bias had greater activity in the left inferior parietal lobule than those whose judgments were not biased. These results not only indicate that group membership sometimes distorts action perception, but also suggest that such distortions are mediated in part by the MNS.

It's also important to note, however, that the degree to which motor resonance is modulated by social categorization can be flexibly altered by manipulating the observer's mindset. This was recently shown by Gutsell et al. (2020) in an EEG study that followed up on a previous investigation by Gutsell and Inzlicht (2010). The subjects had diverse racial/ethnic backgrounds, but none of them were Black. The researchers first showed the subjects a picture of a Black male university student and asked them write a short narrative about him, either remaining as objective as possible or taking his perspective. Then they recorded the subjects' magnitude of mu suppression in response to videos of ball-squeezing actions performed by either non-Black or Black agents. The subjects who adopted an objective mindset displayed significant mu suppression for the non-Black (in-group) agents but not the Black (out-group) ones. In contrast, the subjects who adopted a perspective-taking mindset displayed significant mu suppression for both kinds of agents. These findings suggest that, as Gutsell et al. (2020, p. 50) put it, "when in a perspective-taking mindset, people are more likely to resonate even with out-group members, which could provide them with richer, more intuitive understandings of others' actions and intentions".

5.3. Factor 20: interpersonal liking

The MNS is also modulated by whether an observer has positive or negative feelings toward an actor. In an fMRI study Sobhani et al. (2012) first familiarized their subjects, all of whom were Jewish men, with the biographies of eight individuals, half of whom were portrayed as likable open-minded people, and half of whom were portrayed as dislikable neo-Nazis. Then the researchers scanned the subjects while they watched videos of those eight individuals performing simple arm/hand actions. Although univariate analyses of the imaging data did not reveal any brain regions that were differentially engaged by the sight of actions performed by the two kinds of agents, multivariate pattern analyses disclosed contrasting neural signatures of the two kinds of agents in the right ventral premotor cortex. Thus, it appears that this particular component of the MNS can discriminate between watching a person you like perform an action and watching a person you dislike perform an action.

5.4. Factor 21: eye contact

Finally, several TMS studies have shown that motor resonance is enhanced by direct eye contact between actor and observer. This was first demonstrated by Prinsen et al. (2017), who found that during the observation of simple intransitive hand actions, muscle-specific corticospinal excitability was greater when there was direct rather than averted gaze between the agent and the subjects. In a subsequent TMS study, Prinsen and Alaerts (2019) found that this facilitation of mirror responses by eye contact was especially strong in a live two-person context, compared to when the actions were presented as recorded videos. Furthermore, in a study that employed both TMS and EEG, Prinsen and Alaerts (2020) showed that during action observation eye contact was associated not only with greater corticospinal excitability, but also with greater mu suppression. And in another recent TMS study, Betti et al. (2018) found that when an agent was about to initiate a "social request" gesture, if he looked straight at the observer there was an early increase in corticospinal excitability, but if his gaze was averted there was no anticipatory motor resonance. In sum, this new line of research suggests that gaze cues can significantly influence MNS

responses by either drawing the observer's attention toward the agent's behavior or diverting it away from that behavior (see also Wang et al., 2011a, 2011b).

6. Factor 22: context

Finally, it's worth adding that the MNS is also modulated by the situational context in which actions take place. For instance, when subjects see an agent throw a ball, their hand-related corticospinal excitability is significantly greater when that action is directed at another person (a social context) than when it is directed at a wall (a nonsocial context), even though the kinematics are the same (Bucchioni et al., 2013). Likewise, when subjects see an agent squeeze a sponge, their hand-related corticospinal excitability is significantly greater when that action is performed over a sink (a congruent context) than when it is performed over a cardboard box (an incongruent context), even though the kinematics are the same (Riach et al., 2018).

Furthermore, the observation of contextually odd or inappropriate actions often *inhibits* motor resonance in the relevant parts of the primary motor cortex (Koch et al., 2010; Amoroso et al., 2016; see also several related findings discussed in the subsections on *Factors 15* and *17*). This suppression of precise mirror responses may be driven by higher-level prefrontal regions that recognize the action-context mismatch (Wurm and Schubotz, 2012), and it may occur together with recruitment of the mentalizing system, which is better suited than the MNS to decipher the hidden thoughts and feelings of agents (Brass et al., 2007; see also Van Overwalle and Baetens, 2009).

In addition, when an interactive social context implies that an observer should not covertly imitate an agent's specific action but instead prepare a different complementary one (e.g., to lift a cup after seeing an agent offer to fill it), the observer's MNS activity shifts accordingly (for a review see Sartori and Betti, 2015; see also Betti et al., 2017, which is summarized in the subsection on *Factor 10*).

7. Theoretical implications

Having described 22 factors that modulate the MNS during action observation, it's time to step back from the minutiae and consider their broader ramifications for competing theories of the MNS. Since a thorough evaluation of all the frameworks that have been proposed so far is beyond the scope of this paper, the following discussion is restricted to highlighting the major strengths and weaknesses of four prominent models in terms of how well they can account for the multifarious findings reviewed above. As mentioned in the Introduction, each model is arguably supported by some findings, challenged by others, and relatively unaffected by still others. These assessments are shown in Table 1.

Two important caveats are as follows. First, as we will see, it is sometimes difficult to determine with confidence whether a given model really makes clear predictions about a given factor and hence may be evaluated positively or negatively based on the findings, or if instead it does not actually take a stand regarding that factor and hence is largely impervious to the available data. Different scholars may make different judgments about such ambiguous cases, and for this reason some of the decisions reached here should, as indicated below, be treated as tentative. Second, because the findings for *Factor 4* ("Biomechanical possibility") and the robot-related results for *Factor 7* ("Similarity to humans") are quite variable and hard to interpret, they will not be considered in the assessment of models.

7.1. The Direct Matching Model

7.1.1. Central tenets

The Direct Matching Model (DMM) was developed by Rizzolatti and colleagues and has garnered a great deal of attention. In their initial presentation of this approach, Rizzolatti et al. (2001, p. 661) claimed

that "we understand actions when we map the visual representation of the observed action onto our motor representation of the same action. According to this view, an action is understood when its observation causes the motor system of the observer to 'resonate'. So, when we observe a hand grasping an apple, the same population of neurons that control the execution of grasping movements becomes active in the observer's motor areas".

This characterization of the DMM, however, generated some confusion in the field, prompting [Rizzolatti and Sinigaglia \(2010, p. 269\)](#) to recast it as follows: "In the earlier studies on the mirror mechanism, it was indeed not clearly specified that the parieto-frontal mirror mechanism in humans is involved in two kinds of sensory-motor transformation—one mapping the observed movements onto the observer's own motor representation of those movements (movement mirroring), the other mapping the goal of the observed motor act onto the observer's own motor representation of that motor act (goal mirroring)...."

It's also notable that in a subsequent paper [Rizzolatti et al. \(2014, p.677\)](#) acknowledged the following shortcoming of the DMM: "The weakness is that it is a strictly forward model that does not take into account backward projections, present in large number in the parieto-frontal circuits". Later in the same passage, however, the authors attempt to downplay the significance of the sorts of prior expectations that are usually thought to be mediated by backward projections: "While there is no doubt that the context and the object semantics might be important cues for understanding the intention underlying a given motor act, this is not necessary for understanding the goal of a motor act. Direct activation of parietal and premotor neurons is sufficient for this purpose".

7.1.2. Supportive findings

It makes sense to begin with factors involving the observed action itself. Even though the DMM is concerned more with transitive than intransitive actions, it is supported by most if not all of the data regarding *Factor 1* ("Transitivity"). This is because it posits both (1) goal-mirroring mechanisms, which are relevant not only to transitive actions but also to intransitive ones for which the intention is clear, and (2) movement-mirroring mechanisms, which are relevant to all actions. The DMM is likewise bolstered by almost all of the data discussed in connection with *Factor 2* ("Relative influence of kinematics and goals"). The main exception is [Senna et al.'s \(2014\)](#) finding that the sight of grasping actions performed with a foot enhanced the observers' corticospinal excitability not only for a foot muscle but also for a hand muscle. While the DMM can arguably accommodate this result, it may not have predicted it. The DMM is, however, clearly substantiated by *Factor 3* ("Laws of biological motion"), since it correctly predicts that kinematically natural actions evoke greater MNS responses than kinematically unnatural ones.

Turning to factors involving the actor, further evidence for the DMM comes from those studies regarding *Factor 7* ("Similarity to humans") that compared human actions with animal actions. In particular, the model strongly predicts [Buccino et al.'s \(2004\)](#) and [Amoruso and Urgesi's \(2016\)](#) convergent finding that the MNS is recruited primarily by actions that belong to the onlooker's own motor repertoire. The DMM is also compatible with [Amoruso and Urgesi's \(2016\)](#) additional finding that through observational learning the MNS can be "tuned" to resonate (at the goal level) to some initially unfamiliar animal actions, though the model may not have actually predicted this outcome. The DMM appears to be further reinforced by the data for *Factor 8* ("Number of agents"), especially [Cracco et al.'s \(2019\)](#) finding that the MNS can simultaneously represent two different gestures performed by two different agents. Once again, however, some scholars might maintain that the model does not necessarily predict such responses.

Shifting now to factors involving the observer, the DMM receives additional corroboration from numerous studies regarding *Factor 13* ("Action familiarity and expertise") which have shown that the more experience and skill observers have with a certain type of action, the

more extensively and precisely their MNS is engaged when they see someone else perform it, presumably reflecting their capacity for increasingly accurate motor simulations. The model is also supported by an important finding regarding *Factor 14* ("Learned associations")—specifically, [Ubaldi et al.'s \(2015\)](#) discovery that even after subjects have been trained to link particular types of observed movements with different types of executed movements, they still display normal stimulus-driven mirror responses that occur earlier than the arbitrary rule-based counter-mirror responses. Relatedly, further evidence for the DMM comes from several studies regarding *Factor 15* ("Expectations") which suggest that even when observed actions violate the onlooker's expectations, the MNS sometimes replicates their kinematics with high fidelity, apparently in an automatic bottom-up manner that is implemented by parietofrontal circuits.

Lastly, with respect to factors involving the relationship between actor and observer, one of them—namely, *Factor 18* ("Postural congruence")—seems to provide additional support for the DMM, although the available data are complex, mixed, and based on just two studies. First, the model can arguably account for [Urgesi et al.'s \(2006\)](#) finding that muscle-specific motor resonance is influenced in a biologically constrained way by the postural congruence between actor and observer. Second, the model can also explain most of [Alaerts et al.'s \(2009\)](#) results regarding postural congruence; however, it may not have predicted the rather surprising discovery that in one of their incongruent conditions half the subjects resonated with the perceived movements not according to the intrinsic motor commands, but instead according to the direction in extrinsic space.

7.1.3. Challenging findings

In contrast to the many supportive findings just described, a variety of other results pose difficulties for the DMM. First of all, under the rubric of factors involving the observed action itself, the data regarding *Factor 5* ("Dynamic vs. static stimuli") cannot easily be explained by the model. Even though the MNS appears to respond more robustly to dynamic than static presentations of actions, there is a great deal of evidence that it is in fact engaged by static stimuli. This is problematic for the DMM because understanding static depictions of familiar actions depends on prior knowledge about how such actions unfold over time; however, as indicated above in the summary of this model's central tenets, it does not incorporate such top-down influences, since it only posits feedforward pathways.

Turning to factors involving the actor, the DMM is clearly threatened by several studies regarding *Factor 7* ("Similarity to humans") which showed that the MNS is sensitive to the movements of simple inanimate shapes, like colored circles or triangles. [Ramsey and Hamilton \(2010, p. 2776\)](#) express the basic problem as follows: "a mechanism that matches biological form and motion cannot apply to [these] findings because the shapes that served as actors had neither hand-like body parts nor biological motion trajectories".

As for factors involving the observer, the DMM faces many obstacles. It cannot easily account for some of the results regarding *Factors 11* and *12* ("Perspective" and "Handedness"). Most notably, the model is at odds with [Sartori et al.'s \(2013\)](#) and [Cabinio et al.'s \(2010\)](#) convergent finding that the observer's own hand preference strongly influences MNS responses to perceived actions, regardless of which hand the agent uses. In addition, contrary to the DMM, several studies have shown that the MNS is sometimes recruited more heavily by unfamiliar than familiar actions (*Factor 13*, "Action familiarity and expertise"). The model also precludes the possibility of counter-mirror responses, yet several studies have shown that they can develop through training (*Factor 14*, "Learned associations"). Furthermore, there is considerable evidence that the MNS is sometimes modulated by expectations (*Factor 15*, "Expectations"); however, the DMM cannot explain these findings because, as already mentioned, it does not incorporate top-down influences. For the same reason, it is also contradicted by many striking results which show that the MNS is sensitive to the "subjective value"

that an action has for an observer (*Factor 17*, "Motivation, emotion, personality, and moral judgment").

Finally—and, once again, largely because it is strictly feedforward—the DMM is at odds with numerous findings which show that the MNS is modulated not only by various factors involving the relationship between actor and observer (*Factor 19*, "Racial, ethnic, and abstract social group congruence"; *Factor 20*, "Interpersonal liking"; and *Factor 21*, "Eye contact"), but also by the situation or setting in which an action takes place (*Factor 22*, "Context").

7.1.4. Neutral findings

The DMM does not appear to take a strong stand regarding the following factors: *Factor 6* ("Degree of realism: live vs. recorded, 3D vs. 2D, & point-light stimuli"); *Factor 9* ("Awareness"); *Factor 10* ("Attention"); and *Factor 16* ("Prior social interaction"). Hence, the relevant findings do not seem to bear either positively or negatively on the model.

7.2. The Predictive Coding Model

7.2.1. Central tenets

First proposed by Kilner et al. (2007a, 2007b), the Predictive Coding Model (PCM) is one manifestation of the increasingly popular view that prediction plays a critical role in essentially all forms of perception (e.g., Friston, 2005; Mesulam, 2008; Clark, 2013; Panichello et al., 2013; Summerfield and de Lange, 2014; Feldman Barrett and Simmons, 2015). In general, this view assumes that successive cortical levels in perceptual hierarchies have bidirectional connections that allow reciprocal or recurrent processing interactions. Based largely on past experience and contextual cues, higher levels continuously generate top-down predictions and send them to lower levels, which compare those predictions with bottom-up sensory signals. When mismatches are detected, prediction errors are relayed back up to higher levels, which then make corrections to minimize further prediction errors.

Kilner et al. (2007b, p. 161–2) describe the application of this view to action observation, and more specifically to the MNS, as follows: "For action observation the essence of this approach is that, given a prior expectation about the goal of the person we are observing, we can predict their motor commands. Given their motor commands we can predict the kinematics on the basis of our own action system. The comparison of this predicted kinematics with the observed kinematics generates a prediction error. This prediction error is used to update our representation of the person's motor commands. Similarly, the inferred goals are updated by minimizing the prediction error between the predicted and inferred motor commands. By minimizing the prediction error at all the levels of the MNS, the most likely cause of the action will be inferred at all levels (intention, goal, motor and kinematic). This approach provides a mechanistic account of how responses in the visual and motor systems are organized and explains how the cause of an action can be inferred from its observation".

Before assessing the PCM vis-à-vis the various factors that modulate the MNS, it's important to note that powerful evidence for this model comes from several fMRI studies that are not discussed below because they do not address any of those factors. These studies are worth citing here, however, because they show that, as maintained by the theory, information does in fact flow bidirectionally not only between the visual system and the MNS, but also between the parietal and frontal components of the MNS (Kokal and Keysers, 2010; Schippers and Keysers, 2011; Sasaki et al., 2012).

7.2.2. Supportive findings

Starting with factors involving the action, the PCM is supported by the bulk of results regarding *Factor 1* ("Transitivity"). Perhaps most significantly, it can account quite well for the finding that when intransitive actions have clear intentions, the MNS responses that they evoke are similar to those evoked by transitive actions. In keeping with

the PCM, for both types of actions the observer can use the inferred purpose to predict the motor commands. The model is also well-suited to explain most of the data regarding *Factor 2* ("Relative influence of kinematics and goals"). Like the DMM, however, it would probably not have predicted the results of Senna et al.'s (2014) study. Another virtue of the PCM is that it is substantiated by *Factor 3* ("Laws of biological motion"), since it correctly predicts that the MNS is sensitive to the kinematic regularities of human movements. Moreover, the model receives additional reinforcement from *Factor 5* ("Dynamic vs. static stimuli"). Many studies have shown that the MNS responds to static depictions of familiar actions, and this engagement may reflect the fact that understanding such actions depends on predictive processing derived from previous experience and conceptual knowledge.

Switching to factors involving the actor, the PCM is supported by much of the data for *Factor 7* ("Similarity to humans"). It can easily account for Buccino et al.'s (2004) and Amoruso and Urgesi's (2016) convergent finding that the degree to which the MNS responds to animal actions is contingent upon the degree to which humans can perform them. Moreover, given this model's emphasis on prior expectations about an actor's goals, it can also provide a reasonable explanation for Amoruso and Urgesi's (2016) additional discovery that as people become more familiar with the purposes of animal actions, their MNS responses become more robust. Of special interest, however, is that the PCM seems to be further reinforced by those studies which have shown that the MNS is even engaged by the movements of circles and triangles. This is because, as Engel et al. (2008b) and Ramsey and Hamilton (2010) point out, such MNS responses may reflect the top-down attribution of goals to such inanimate stimuli, as a heuristic strategy for understanding them. In this context, it's also worth reiterating that, as mentioned in the subsection on *Factor 7*, Schubotz (2007) argues that the premotor cortex, which is a major component of the MNS, contributes to the prediction of observed events that cannot be reproduced by humans—a proposal that is clearly consistent with the PCM.

Shifting to factors involving the observer, evidence for the PCM comes from many studies regarding *Factor 13* ("Action familiarity and expertise"). As indicated above, the model maintains that during action observation the MNS constantly compares bottom-up visual representations with top-down expectations based on prior knowledge and the current context, with the aim of minimizing prediction error. From this theoretical perspective, when people see actions for which they have a high level of familiarity/expertise, their strong MNS responses may be due to very precise predictions that project posteriorly, whereas when people see actions for which they lack a high level of familiarity/expertise, their strong MNS responses may be due to many prediction error messages that project anteriorly. Now, an interesting question that follows from this line of thinking is what happens to the MNS when people see actions for which they have an intermediate level of familiarity/expertise. According to the PCM, such stimuli should only engage the MNS to a moderate degree, and it's notable that if this were the case, it would support a nonlinear rather than a linear effect of familiarity/expertise on the MNS. This hypothesis has recently been tested in a few studies, but only some of the results fit the predicted kind of nonlinear effect (Liew et al., 2013; Gardner et al., 2017; for a different kind of nonlinear effect see Chen et al., 2020).

Returning to factors involving the observer, further support for the PCM comes from some of the data regarding *Factor 15* ("Expectations"). In particular, a number of studies have shown that, in keeping with the model, the MNS is often influenced by the onlooker's expectations about the perceived action. For instance, while discussing their own intriguing discovery—namely, that during the observation of grasping actions, muscle-specific motor resonance only occurred when the agent's hand configuration matched the shape of the target object—Craighero et al. (2014, p. 667) point out that "motor resonance is an active process considering kinematics, context cues, and past motor experience to predict others' action outcomes ... and to check the appropriateness of the observed action with respect to the predicted goal".

Finally, the PCM is strongly corroborated by the results regarding *Factor 22* ("Context"). One of the model's basic claims is that observers use the situations in which agents behave to constrain their inferences about those agents' goals and intentions, and a variety of studies have confirmed that the MNS is in fact very sensitive to such contextual cues.

7.2.3. Challenging findings

The major threat to the PCM comes from some of the data regarding *Factor 15* ("Expectations"). Specifically, several studies have demonstrated that during action observation, if the agent's kinematics differ from what the onlooker predicted, the onlooker's MNS responses sometimes continue to track the actual movements. These findings indicate that, contrary to the PCM, motor resonance is not always influenced by top-down expectations, but is instead sometimes driven primarily by bottom-up signals.

7.2.4. Neutral findings

The PCM does not appear to take a clear position regarding the following factors: *Factor 6* ("Degree of realism: live vs. recorded, 3D vs. 2D, & point-light stimuli"); *Factor 8* ("Number of agents"); *Factor 9* ("Awareness"); *Factor 10* ("Attention"); *Factor 11* ("Perspective"); *Factor 12* ("Handedness"); *Factor 14* ("Learned associations"); *Factor 16* ("Prior social interaction"); *Factor 17* ("Motivation, emotion, personality, and moral judgment"); *Factor 18* ("Postural congruence"); *Factor 19* ("Racial, ethnic, and abstract social group congruence"); *Factor 20* ("Interpersonal liking"); and *Factor 21* ("Eye contact").

Now, for a few of these factors—especially 17, 19, 20, and 21—some scholars might argue that the PCM does make predictions and is actually supported by some of the data. Take, for instance, *Factor 17*. One could contend that the model is bolstered by Liuzza et al.'s (2015) and Craighero and Mele's (2018) convergent finding that motor resonance is suppressed during the observation of immoral actions. In fact, Craighero and Mele (2018) explicitly adopt a version of the PCM to explain their result, arguing that the suppression of motor resonance is driven in a top-down manner by, first, the onlooker's judgment of the agent's action as bad, and second, the onlooker's desire to refrain from covertly replicating that action. This interpretation presupposes, however, that value and valence are inherent parts of the PCM, and such an assumption may not be warranted. Arguments for the relevance of *Factors 19, 20, and 21* to the PCM may also hinge on this potentially unwarranted assumption. It's important to note, though, that one of the main motivations for the Value-Driven Model, which is discussed next, was to augment the PCM by incorporating neurocognitive systems that do process value and valence.

7.3. The Value-Driven Model

7.3.1. Central tenets

As just stated, the Value-Driven Model was deliberately designed to be an extension of the PCM (Aziz-Zadeh et al., 2018). Its distinguishing feature is that it capitalizes on evidence that the MNS receives input from brain regions that subserve reward processing (such as the basal ganglia) and emotion/salience processing (such as the amygdala, anterior insula, and anterior cingulate cortex). According to the VDM, these inputs make the MNS highly sensitive to the sight of actions that have value for the observer, where value is defined as "the subjective importance, worth, or usefulness of something" (Aziz-Zadeh et al., 2018, p. 3). More precisely, "the value of the observed action and actor to the self ... depends on goals, relationships, personal history, context, environment and motivational state" (Aziz-Zadeh et al., 2018, p. 5).

7.3.2. Supportive findings

Because the VDM builds on the basic architecture of the PCM, it inherits all of that model's strengths. What it adds is the ability to explain most if not all of the findings for *Factor 17* ("Motivation, emotion, personality, and moral judgment"), *Factor 19* ("Racial, ethnic, and abstract

social group congruence"), *Factor 20* ("Interpersonal liking"), and *Factor 21* ("Eye contact").

Interestingly, the VDM also seems to be bolstered by many of the results regarding *Factor 6* ("Degree of realism: live vs. recorded, 3D vs. 2D, & point-light stimuli"). Several studies have shown that MNS responses are enhanced when actions are witnessed "in the here and now," and from the perspective of the model, this may reflect the greater meaningfulness of such actions, relative to those that are merely recorded. Similarly, there is some evidence that 3D actions evoke stronger MNS responses than 2D actions, and again the more realistic nature of the former stimuli may increase their salience and significance for viewers, in keeping with the VDM.

7.3.3. Challenging findings

At the same time, the VDM shares the primary weakness of the PCM, which is difficulty accounting for the fact that the MNS is sometimes driven more by how perceived actions actually unfold than by how they are predicted to unfold (*Factor 15*, "Expectations"). Another limitation of the VDM is that its core construct, namely value, is inherently subjective and hence often hard to measure.

7.3.4. Neutral findings

As with the PCM, the VDM does not appear to be affected by the data regarding the following factors: *Factor 8* ("Number of agents"); *Factor 9* ("Awareness"); *Factor 10* ("Attention"); *Factor 11* ("Perspective"); *Factor 12* ("Handedness"); *Factor 14* ("Learned associations"); *Factor 16* ("Prior social interaction"). One notable exception, however, involves *Factor 11* ("Perspective"). As Aziz-Zadeh et al. (2018, pp. 5–6) point out, the VDM correctly predicts that "socially relevant stimuli, like individuals facing toward the observer rather than away from the observer, should show modulation in the MNS (Kilner et al., 2007a, 2007b)". Most of the other findings for *Factor 11*, however, do not seem to bear directly on the model.

7.4. The Associative Model

7.4.1. Central tenets

The fourth framework to be considered here is the Associative Model (AM), which states that mirror neurons emerge ontogenetically by means of the same kind of domain-general learning that underlies Pavlovian conditioning (Heyes, 2010; Cook et al., 2014; Catmur et al., 2016). In short, the basic claim is that mirror neurons are forged through correlated experiences of action observation and execution. According to the AM, such associations between seeing and doing occur at all levels of action representation, from fine-grained motor commands to overarching intentions. Moreover, this kind of associative learning depends not only on contiguity (i.e., events occurring close together in space and time), but also on contingency (i.e., the degree to which one event reliably predicts another), and there is no requirement that the contingencies between action observation and execution be for the very same type of action; rather, it is possible to develop mirror neurons that capture contingencies between observing one type of action (e.g., releasing an object) and executing another type of action (e.g., grasping an object). In addition, the AM maintains that the MNS is susceptible to contextual modulation, including influences from the situation or setting in which actions are observed. As Catmur et al. (2016, p. 526) point out, this is based on findings from the literature on conditioning: "if a stimulus is associated with two responses, each in a different context, then the context determines which association, representing a response-outcome contingency, is cued by the stimulus". Finally, the AM is "functionally permissive" (Cook et al., 2014, p. 178) insofar as it allows, but does not assume, that the MNS contributes to action understanding and other forms of social cognition.

7.4.2. Supportive findings

Because the AM assumes that visuomotor mappings develop for all

kinds of actions and at all levels of representation, it is supported by the data for *Factors 1* and *2* ("Transitivity" and "Relative influence of kinematics and goals"). This model can even account quite nicely for [Senna et al.'s \(2014\)](#) discovery that the sight of an agent grasping something with his foot enhanced the observers' corticospinal excitability of a hand muscle. From the perspective of the theory, such motor resonance most likely reflects an associative process of generalization from the perceived atypical form of grasping to the simulated typical form. The AM is also bolstered by *Factor 3* ("Laws of biological motion"), since it correctly predicts that kinematically natural actions evoke greater MNS responses than kinematically unnatural ones. In addition, the AM receives further support from *Factor 5* ("Dynamic vs. static stimuli"), since the well-established responsiveness of the MNS to mere snapshots of actions may reflect the contingent or predictive association between one phase of an action and subsequent phases.

Turning to factors involving the actor, the AM can provide compelling explanations for many of the findings regarding *Factor 7* ("Similarity to humans"). For one thing, it makes sense that as the ability of humans to perform animal actions decreases, so does the sensitivity of the MNS to those actions, since the pertinent visuomotor mappings get weaker. And yet, the AM can also account for [Amoruso and Urgesi's \(2016, p. 770\)](#) finding that through extensive exposure the MNS can be "tuned" to resonate to initially unfamiliar animal actions. As the researchers point out, "...attribution of meaning to the other species' impossible actions (e.g. moving the tail) might have been accomplished through a generalization process; in some ways, moving the tail serves the same communication functions as moving the arm in a human greeting.... Furthermore ... the differential pattern of activity in experienced and non-experienced individuals must be ascribed to the extent of visual familiarity with the actions and their associations with corresponding motor plans" ([Amoruso and Urgesi, 2016, p. 770](#)). It's also noteworthy that the AM can account rather well for the finding that the MNS is even engaged by the movements of circles and triangles. Due to their rudimentary visual resemblance to human actions, such stimuli may ignite associated motor programs. As [Engel et al. \(2008b, p. 2040\)](#) put it, "a relation to movement execution can be constructed for non-biological stimuli ... if some visual features trigger strong associations with human movements" (see also [Press and Cook, 2015](#)). Lastly, the AM seems to be supported by the data for *Factor 8* ("Number of agents"), including [Cracco et al.'s \(2019\)](#) finding that the MNS can simultaneously represent two different gestures performed by two different agents. The model arguably predicts such parallel visuomotor mappings, although a skeptic might disagree.

As for factors involving the observer, the AM appears to be bolstered by some of the results regarding *Factor 12* ("Handedness"). Specifically, if the model assumes that the MNS captures observation-execution associations that are biased toward the observer's own motor preferences, then it correctly predicts the following findings from [Cabinio et al. \(2010\)](#): first, right-handers' MNS responses were always relatively left-lateralized, but more so for observed right-hand than left-hand actions; and second, strong left-handers' MNS responses were always relatively right-lateralized, but more so for left-hand than right-hand actions (see also [Sartori et al., 2013](#)). The AM is also corroborated by those studies regarding *Factor 13* ("Action familiarity and expertise") that have revealed a positive correlation between the magnitude of MNS responses to observed actions and the onlooker's skill at executing those actions. Another virtue of the AM—one that warrants special emphasis—is that it correctly predicts the capacity of the MNS to develop counter-mirror responses to observed actions, as demonstrated by several studies regarding *Factor 14* ("Learned associations"). What's more, the model is further reinforced by a number of studies regarding *Factor 15* ("Expectations") which have shown that the MNS is often influenced by the onlooker's predictions about perceived actions. From the perspective of the AM, however, such predictions do not necessarily reflect top-down processing; instead, they simply reflect learned contingencies.

Finally, the AM is strongly supported by the data for *Factor 22* ("Context") because, as mentioned above, it not only allows for but actually anticipates the contextual modulation of the MNS. Importantly, the model is particularly well-suited to elucidate several studies which have shown that when a social situation implies that an observer should not covertly imitate an agent's specific action but instead prepare a different complementary one (e.g., to lift a cup after seeing an agent offer to fill it), the observer's MNS represents the appropriate response rather than the inappropriate one. In their review of this literature, [Sartori and Betti \(2015, p. 5\)](#) point out that "the relation between observed and executed actions seems then to be coordinated by a social associative memory which apparently matches some actions to their natural social responses regardless of who is actually performing the action...." And they go on to note that, in accord with the AM, there may be "no difference between congruent [i.e., mirror], incongruent [i.e., counter-mirror], and complementary responses, as long as they have been associatively linked".

7.4.3. Challenging findings

Despite the strengths just described, the AM also has some significant weaknesses. It cannot easily explain several studies which have shown that the MNS is sometimes recruited more heavily by unfamiliar than familiar actions (*Factor 13*, "Action familiarity and expertise"). In addition, it is threatened by [Ubaldi et al.'s \(2015\)](#) finding that even after subjects have been trained to link certain types of observed movements with different types of executed movements, they still display normal stimulus-driven mirror responses that occur earlier than the arbitrary rule-based counter-mirror responses (*Factor 14*, "Learned associations"). And lastly, it has trouble accommodating the evidence that during action observation, if the agent's kinematics differ from what the onlooker predicted (on the basis of learned contingencies), the onlooker's MNS responses sometimes continue to mimic the actual movements rather than reflecting the expected ones (*Factor 15*, "Expectations").

7.4.4. Neutral findings

The AM does not appear to take a strong stand regarding the following factors: *Factor 6* ("Degree of realism: live vs. recorded, 3D vs. 2D, & point-light stimuli"); *Factor 9* ("Awareness"); *Factor 10* ("Attention"); *Factor 11* ("Perspective"); *Factor 16* ("Prior social interaction"); *Factor 17* ("Motivation, emotion, personality, and moral judgment"); *Factor 18* ("Postural congruence"); *Factor 19* ("Racial, ethnic, and abstract social group congruence"); *Factor 20* ("Interpersonal liking"); and *Factor 21* ("Eye contact"). Hence, the relevant findings do not seem to affect the model either positively or negatively.

8. Conclusion

During the past 25 years, a tremendous amount of empirical research has explored how the MNS contributes to the perception and understanding of actions. These investigations have shown that when the MNS responds to observed actions, it does so in a highly variable manner that is influenced by myriad factors. The first aim of this paper has been to provide the most comprehensive survey to date of these modulatory factors. To that end, 22 distinct factors have been described, broken down into the following sets: six involving the action; two involving the actor; nine involving the observer; four involving the relationship between actor and observer; and one involving the context. The second aim has been to evaluate the implications of these findings for four prominent theoretical models of the MNS: the Direct Matching Model; the Predictive Coding Model; the Value-Driven Model; and the Associative Model. These assessments revealed that although each model is supported by a wide range of findings, each one is also challenged by other findings and relatively unaffected by still others. Thus, an important direction for future work should be to develop a more satisfactory model of the MNS, one that not only comes closer to meeting what [Keas \(2018\)](#) calls the major evidential virtues of good theories

(evidential accuracy, causal adequacy, and explanatory depth), but also strives to achieve the other virtues as well: coherentist (internal consistency, internal coherence, and universal coherence); aesthetic (beauty, simplicity, and unification); and diachronic (durability, fruitfulness, and applicability).

Appendix A. The Peer Review Overview and Supplementary data

The Peer Review Overview and Supplementary data associated with this article can be found in the online version, at doi:<https://doi.org/10.1016/j.pneurobio.2021.102128>.

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